Supervised Nonlinear Hyperspectral Unmixing With Automatic Shadow Compensation Using Multiswarm Particle Swarm Optimization

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***Abstract*— The presence of shadows has always been a troublesome problem in image processing and can also affect spectral unmixing with hyperspectral remote sensing images. Traditional unmixing algorithms regard shadows as a special type of ground cover, so they can only estimate real materials' sunlit abundances, and materials with low reflectance may be wrongly recognized as shadows. Without regarding shadows as ground cover, we propose a supervised nonlinear unmixing method to accurately estimate real materials' total abundances inside and outside shadow areas. First, sunlit and shadowed constituents in every pixel of an image are modeled explicitly and integrated by a tractable bilinear mixing mechanism. Second, based on the constructed model, the strong sparsity of shadow spatial distribution and the spatial correlation among neighboring material abundances are exploited to produce a constrained optimization problem for nonlinear unmixing. Third, an existing unmixing framework based on particle swarm optimization is extended to calculate unknown variables of the optimization problem. Three alternatingly updated swarms using improved dimensional division strategies are designed to accordingly address the unmixing subproblems with respect to variables to be estimated during the solution search. This process has the potential to be generalized to solve complex nonlinear unmixing optimization problems. Finally, model-based simulated data, virtual citrus orchard data, and real hyperspectral remote sensing images are used in experiments to evaluate the proposed method and compare it with traditional and state-of-the-art nonlinear unmixing algorithms. Experimental results verify that the proposed method can achieve acceptable unmixing performance when managing nonlinear mixing effects and shadows.**

***Index Terms*— Divide-and-conquer particle swarm optimization (PSO), hyperspectral imagery, multiple swarms, shadow compensation, supervised nonlinear spectral unmixing.**

NOMENCLATURE

A. Symbols of Models and Optimization Problems

$\mathbb{R}^{n\times m}$	$n \times m$ real matrix, and r, n, and m are the number of endmembers, bands, and pixels, respectively.
$\mathbf{1}$, $\mathbf{0}$	Column vectors of all 1s and 0s, respectively.
$\boldsymbol{x}_j$, $\hat{\boldsymbol{x}}_j$	jth pixel column vector and its corresponding reconstructed pixel, respectively.
$\boldsymbol{x}_j^{1\text{st}}$, $\boldsymbol{x}_j^{2\text{nd}}$	Single scattering term and second-order scattering term of the jth pixel, respectively.
$\boldsymbol{x}_j^{\text{sunlit}}$, $\boldsymbol{x}_j^{\text{shadow}}$	Vectors of the jth pixel's sunlit part and shadowed part, respectively.
$\boldsymbol{X}$, $\boldsymbol{X}_{k,:}$	Hyperspectral data matrix and the kth row vector of $\boldsymbol{X}$, respectively.
$\boldsymbol{a}_i$, $\boldsymbol{A}$	ith endmember vector and an endmember matrix, respectively.
$\boldsymbol{s}_j$, $\hat{\boldsymbol{s}}_j$, $\boldsymbol{S}$	Abundance vector of the jth pixel, $\boldsymbol{s}_j$'s estimated abundance vector, and an abundance matrix, respectively.
$\boldsymbol{s}_j^v$, $\boldsymbol{s}_j^h$	Abundance vectors of the pixels neighboring to the jth pixel in the vertical and the horizontal directions, respectively.
$\boldsymbol{\varepsilon}_j$	Noise vector.
$\gamma_{i,k,j}$, ξ_j	Nonlinear parameters of GBM and PPNM, respectively.
$\underline{P}_j$, Q_j	MLM's nonlinear probability parameter and SMLM's shadow parameter, respectively.
$\odot$	Hadamard product.
$\|\cdot\|_{1,1}$, $\|\cdot\|_{\text{TV}}$, $\|\cdot\|$	ℓ_1 norm, TV regularizer, and absolute value operator, respectively.
$\boldsymbol{I}_{\text{direct}}$, $\boldsymbol{I}_{\text{indirect}}$, $\boldsymbol{R}_{\text{reference}}$	Direct irradiance, indirect irradiance, and radiance of a referenced calibration panel, respectively.
$\boldsymbol{h}_j$, $\boldsymbol{\Delta}_j$	Sunlit and shadowed coefficient vectors of the jth pixel, respectively.
$h_{i,j}$, $d_{i,j}$	Sunlit coefficient of the ith endmember in $\boldsymbol{h}_j$ and shadowed coefficient of the ith endmember in $\boldsymbol{\Delta}_j$, respectively.
$\widehat{d}_j$	Summation of all the elements in $\boldsymbol{\Delta}_j$.
$\boldsymbol{c}$	SCBMM's wavelength-dependent scaling vector.
λ_1, λ_2	Regularization parameters.

B. Symbols of Multiple Swarm PSO

M, t, $t_{\max}$	Swarm's size, iteration number, and maximum iteration, respectively.
$\boldsymbol{P}_i^{(t)}$, $\boldsymbol{V}_i^{(t)}$	ith particle's position and velocity of a PSO's swarm in iteration t, respectively.

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$\boldsymbol{P}_{\text{ipbest}}^{(t)}$, $\boldsymbol{P}_{\text{gbest}}^{(t)}$	ith particle's best position in history and the swarm's global best position, respectively.
ω, $\underline{c}_1$, $\underline{c}_2$	Inertia weight and two acceleration coefficients, respectively.
Swarm$_{\boldsymbol{S}}$, Swarm$_{\boldsymbol{\Delta}}$, Swarm$_{\boldsymbol{c}}$	Three swarms for optimizing $\boldsymbol{S}$, $\boldsymbol{\Delta}$, and $\boldsymbol{c}$, respectively.
$\boldsymbol{P}_{\boldsymbol{S}_\text{gbest}}$, $\boldsymbol{P}_{\boldsymbol{\Delta}_\text{gbest}}$, $\boldsymbol{P}_{\boldsymbol{c}_\text{gbest}}$	Swarm's global best position of Swarm$_{\boldsymbol{S}}$, Swarm$_{\boldsymbol{\Delta}}$, and Swarm$_{\boldsymbol{c}}$, respectively.
$\boldsymbol{P}_{\text{pbest_slice_}j}^{(t)}$, $\boldsymbol{P}_{\text{gbest_slice_}j}^{(t)}$	jth subcomponent slice of $\boldsymbol{P}_{\text{pbest}}^{(t)}$ and $\boldsymbol{P}_{\text{gbest}}^{(t)}$, respectively.
$\boldsymbol{P}_{i_\text{slice_}j}^{(t)}$	jth subcomponent slice of $\boldsymbol{P}_i^{(t)}$. The jth value of $\boldsymbol{P}_i^{(t)}$'s fitness vector is the best in a swarm.
f_{SCBMM}	Reconstruction error based on SCBMM.
$f_{\text{SCBMM}}^{\text{C}j}$, $f_{SCBMM}^{\text{R}k}$	f_{SCBMM} 'sjth subcomponent obtained through pixel-based division and f_{SCBMM}'s kth subcomponent obtained through band-based division, respectively.
$\boldsymbol{fitness}_{\boldsymbol{S}}$, $\boldsymbol{fitness}_{\boldsymbol{\Delta}}$, $\boldsymbol{fitness}_{\boldsymbol{c}}$	Vectorized fitness functions of Swarm$_{\boldsymbol{S}}$, Swarm$_{\boldsymbol{\Delta}}$, and Swarm$_{\boldsymbol{c}}$, respectively.
g_j, z_j	jth subcomponent of $\ell_1$norm and TV regularizer based on the pixel-based division, respectively.
$r_1^{(t)}$, $r_2^{(t)}$, rand$_{0-1}$	Random numbers uniformly drawn in the set (0, 1).
$\boldsymbol{B}_{\min}$, $\boldsymbol{B}_{\max}$	Lower and upper bound matrices of particles.

I. Introduction

ABUNDANT spectral information in a series of continuous bands provided by hyperspectral imagery can markedly enhance ground-object recognition during Earth remote sensing observations. The reliable quantitative analysis of inherent properties for materials of interest has become promising [1], [2]. However, complex natural surface conditions at different spatial observation scales always produce mixed pixels in hyperspectral images. In most remote sensing applications, such as environmental monitoring and mineral exploration, mixed pixels may seriously reduce the accuracy of traditional pixel-level information extraction and analysis. As a spatial downscaling technique, spectral unmixing decomposes every pixel into *endmembers* and *abundances* at the subpixel level [1]. Endmembers represent various typical pure ground cover spectra, and abundances correspond to their fractional proportions in the pixels.

Linear and nonlinear mixture models (LMM and NMM) typically serve as a basis for spectral unmixing. In terms of the mixing process's expression, the LMM simplifies light's interactions among different ground covers to be only single scatterings (i.e., pixels are assumed to be linearly constituted by endmembers), while the NMMs' central assumption lies on the nonnegligible multiple scatterings [3], [4]. Thus, in contrast to linear spectral unmixing methods, nonlinear spectral unmixing methods can explain intrinsic nonlinear mixing effects in typical natural scenarios covered by vegetation, urban buildings, or minerals, where so-called complex multilayered or intimate mixtures are distributed [4]. In general, nonlinear unmixing has a strong potential to generate more reasonable unmixing results, particularly for complex natural scenarios, although its corresponding optimization problem is often more difficult to solve than linear unmixing.

In recent years, bilinear mixture models (BMMs) have promoted the study of nonlinear unmixing methods to a certain degree due to their tractable mathematical forms. By incorporating only second-order scatterings that are in the form of the Hadamard products of every two endmember vectors and abundances into the LMM, BMMs can effectively approximate the major contributions of nonlinear mixing residuals [5]. Among many representative BMMs [6]–[10], the generalized bilinear model (GBM) [9] and the polynomial postnonlinear model (PPNM) [10] are popular for nonlinear unmixing due to their good generalization. Other NMMs, such as the p-order polynomial model [11] and the multilinear mixing (MLM) model [12], can also consider high-order scatterings in the complex urban environment. The Hapke model [13] describes nonlinearities via strict physical radiative transfer (RT) equations. Kernel function-based models [14], [15] provide a flexible way to describe different nonlinear mixing effects.

Some recent studies [16]–[19] performed a set of simulated and *in situ* experiments to quantitively compare and validate several BMMs optimized by traditional numerical methods. Due to complex nonlinearities, most current nonlinear unmixing methods focus on abundance estimation, which is also this study's primary focus and highlights that BMM-based abundance estimation commonly performs better than supervised linear unmixing methods, such as fully constrained least squares (FCLS) [20] for typical vegetated scenarios. Based on existing experimental results, efforts have been devoted to more research, including the exploitation of hyperspectral data's nonlinear manifold [21], the reduction of collinearity's negative influence on BMM-based unmixing [22], and the interpretation of wavelength-dependent nonlinearity [23], [24]. The construction of reasonable NMMs and the algorithm design to calculate correct and reliable unmixing results have become two major challenges in the field of model-based nonlinear unmixing.

Illumination variations induced by the local topography of real observed scenarios are often ignored by most BMMs, which are one factor that leads to endmember spectral variability [25]. A typical case is that shadows are generated when direct sunlight is blocked, which often occurs in scenarios such as vegetation-covered areas with large 3-D structures where incident light may transmit many times between the ground and canopy [26]–[28]. Self-shadows or cast shadows [28] appear in different optical processes (see Fig. 1). Although the

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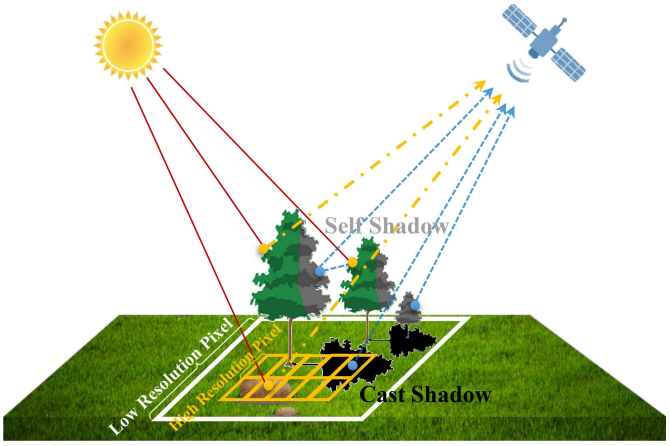


Fig. 1. Multiple scatterings and shadows in the observed pixels.

inherent optical properties of ground covers are illumination-invariant, the apparent reflectance spectra of shadowed pixels derived from atmospheric correction approaches, such as the empirical line method (ELM), are always lower than those of their corresponding sunlit pixels.

Due to the absence of direct irradiance, a large percentage of irradiance received by the shadows comes from the diffused skylight and the light reflected by surrounding objects [27], [28]. Therefore, shadowed pixels appear darker than other sunlit pixels, and their spectral magnitudes and shapes also change significantly with wavelength. Pixels may be fully covered by shadows or contain partial shadows depending on the spatial resolution. Also, the presence of shadows could cause a loss of important information and deteriorate the estimation of critical physical parameters, such as vegetation indices [29].

The issue of shadows has drawn much attention in processing images with high spatial resolution in recent decades. Instead of categorizing shadows as a land cover type, shadow detection and removal methods aim to reduce the impact of shadows as much as possible and produce normally illuminated images, which have always been challenging tasks in computer vision [30]. Thresholding methods, spectral indices, or color invariant methods use a combination of some carefully selected spectral bands for shadow detection [31]–[33]. LiDAR data can be used to generate digital surface models (DSMs) and detect shadows according to geometric structures [34], [35]. Deep artificial neural networks have also recently shown great potential in dealing with shadows [36]–[38].

Compared to remote sensing images with high spatial resolution, the negative effect of shadows on extracting real abundances from hyperspectral images with low spatial resolution at the subpixel level is a challenging issue, and the shadows do not disappear although they may not be observed directly by human eyes in this situation. Current evidence suggests that the shadow contribution in real hyperspectral remote sensing images may be strong in some regions of interest (see Section IV) and cannot be regarded as rare outliers, particularly in quantitative applications, such as the estimation of vegetation biological parameters and impervious surfaces. Conversely, it can be inferred that the frequent appearance of shadows always implies that objects with various large 3-D geometric structures (e.g., trees and artificial buildings) are likely to exist widely in many scenarios. Thus, nonlinear mixing effects cannot be ignored. However, because traditional BMMs are built based on the description of sunlit pixels, shadows are typically regarded as a special type of endmember in spectral unmixing [39]. Real material abundances inside the shadows cannot be described, and the use of shadow endmembers may produce errors in the unmixing processes of sunlit pixels. Hence, it is meaningful to consider both the nonlinear mixing effects and shadows when developing an NMM.

In fact, shadows and the unmixing processes are closely related, and shadows could be removed via unmixing [40]–[45]. For example, the negative effects of endmembers' spectral magnitude variation on abundance estimation caused by shadows could be reduced using spectral angle measures in linear unmixing [40]. Liu *et al.* [41] modeled shadowed endmembers with scaled sunlit endmembers and offsets under the assumption of LMM. The obtained unmixing results were used to remove cloud shadows. Recently, shadows have also attracted much attention in hyperspectral nonlinear unmixing. In the description of shadows, physical RT equations are often used to analyze the changes in radiances and reflectances of the shadowed pixels [42]–[44]. From this perspective, Uezato *et al.* [43] derived an illumination invariant nonlinear unmixing model by combining radiance data and the DSMs generated by LiDAR data. Zhang *et al.* [44] and Zhao *et al.* [45] selected both sunlit and shadowed endmembers for unmixing. Based on the NMMs, the shadowed endmembers were used to estimate the abundances of predetermined shadowed pixels. Thus, sunlit images could be restored using the sunlit endmembers and abundances. Heylen *et al.* [46] proposed a shadow MLM (SMLM) model that used a shadow parameter to balance the contributions of the MLM and the first-order scatterings in nonlinear unmixing. Specifically, a shadowed pixel could be modeled by removing the linear mixing part from the MLM.

These methods have provided valuable explorations for managing shadows in spectral unmixing. However, some of their drawbacks cannot be neglected, such as the requirement of matched auxiliary data and individually separated unmixing processes for sunlit and shadowed pixels. Therefore, errors are likely to propagate among the supplementary procedures and unmixing chains in practical applications. This fact also makes it desirable to develop a robust optimization approach for solving complex models where shadows and nonlinear mixing effects are considered simultaneously. In contrast to frequently used methods, such as the gradient descent algorithm (GDA), computational intelligence approaches, including particle swarm optimization (PSO) [47], [48], are more suitable to manage such nonconvex nonlinear unmixing problems. PSO searches for optimal solutions directly according to particles' fitness functions without gradient calculations, which makes it easier to extend to different nonlinear models. Better local optima are typically used due to the PSO's

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multidirectional stochastic search strategies [48], and PSO's powerful optimization ability has been validated in many related studies that focus on band selection [49], endmember extraction [50]–[53], and unsupervised spectral unmixing [54].

To the best of our knowledge, it is challenging and necessary to further cope with the shadows in nonlinear unmixing in three aspects. In terms of fundamental model construction, a concise NMM that considers shadows may be proposed without selecting shadowed endmembers or auxiliary profiles. In terms of the design for an acceptable unmixing optimization problem, intrinsic features of the shadows and abundances should be used effectively. Finally, in terms of solving complex nonlinear problems, the robust structure of PSO can be reconstructed and improved accordingly to produce accurate and stable unmixing results.

In this study, a supervised nonlinear unmixing method with automatic shadow compensation [27] is proposed. First, each endmember is allowed to have sunlit and shadowed proportional coefficients in every pixel. A pixel's sunlit part is modeled based on the sunlit coefficients and a traditional BMM, while its shadowed part is approximate to the scaled combination of endmembers and their shadowed coefficients. The integration of these two parts and their mutual second-order scatterings produces a shadow-compensated BMM (SCBMM). Second, total variation (TV) and sparse regularizers are assigned for the sunlit and shadowed coefficients in the SCBMM-based unmixing problem to reformulate the original ill-posed problem and avoid overfitting, respectively. Finally, based on the idea of an existing double swarm PSO method [54] in a previous study, we extend this method to nonlinear unmixing by improving the PSO structures according to the reformulated optimization problem. Three swarms whose fitness functions are vectorized by taking advantage of the separability of the SCBMM and regularizers are built to estimate unknown variables alternately.

The major contributions of this article can be briefly summarized as follows.

1) *Mixture Model:* An improved BMM where shadows are not regarded as a real land cover is proposed to model sunlit areas and shadowed areas inside every pixel simultaneously. In the two subpixel-level areas, spectral mixing processes are separately defined, and interactions between sunlit and shadowed materials are explicitly built. Real ground covers' total abundances can be obtained by estimating their sunlit and shadowed coefficients. Traditional abundance loss in the shadows can be compensated after unmixing.
2) *Unmixing Problem:* The nonlinear unmixing optimization problem is formulated by combing the proposed model with two physical constraints based on the sparsity of shadowed coefficients and the spatial similarity of abundances. In this sense, the ill-posed mathematical problem induced by the proposed model's complexity and numerous unknown parameters can be well alleviated, and physically meaningful unmixing results are more likely to be produced.
3) *Optimization Algorithm:* An existing PSO framework is extended to solve the formulated unmixing problem. Three swarms with newly designed dimensional division strategies and a dynamic boundary control strategy are designed to estimate unknown variables in the proposed model, which has satisfactory search accuracy in dealing with high-dimensional problems. This optimization algorithm is user-friendly and can be easily generalized to deal with other complex nonlinear unmixing problems.

The remainder of this article is organized as follows. In Section II, several typical mixture models and the basic global PSO are introduced. Section III describes the proposed SCBMM, the constrained optimization problem for nonlinear unmixing, and a divide-and-conquer PSO method with multiple swarms. Experiments with model-based simulated data, a virtual citrus orchard dataset, and two real hyperspectral images are presented in Section IV. Finally, conclusions and suggestions for future work are summarized in Section V.

The notations used throughout this article are briefly shown according to two categories given in Nomenclature.

II. Related Works

A. LMM and BMMs

With the assumption of the LMM, every pixel is a convex combination of endmembers and abundances [1]. BMMs such as the GBM [9] and the PPNM [10] add second-order scatterings between every two endmembers to the LMM. Let $\boldsymbol{X} \in \mathbb{R}^{n\times m}$ denote the matrix of a hyperspectral dataset, and the jth pixel $\boldsymbol{x}_j \in \mathbb{R}^{n\times 1}$ $(j = 1, 2, \ldots, m)$ can be expressed as (1) and (2) according to the LMM and the BMMs, respectively,

$$\boldsymbol{x}_j = \sum_{i=1}^{r} \boldsymbol{a}_i s_{i,j} + \boldsymbol{\varepsilon}_j = \boldsymbol{x}_j^{1\text{st}} + \boldsymbol{\varepsilon}_j \tag{1}$$

$$\boldsymbol{x}_j = \boldsymbol{x}_j^{1\text{st}} + \boldsymbol{x}_j^{2\text{nd}} + \boldsymbol{\varepsilon}_j \tag{2}$$

$$\boldsymbol{x}_j^{2\text{nd}} = \begin{cases} \sum_{i=1}^{r-1} \sum_{k=i+1}^{r} \gamma_{i,k,j} (\boldsymbol{a}_i \odot \boldsymbol{a}_k) s_{i,j} s_{k,j}, & \text{GBM} \\ \xi_j \sum_{i=1}^{r} \sum_{k=1}^{r} (\boldsymbol{a}_i \odot \boldsymbol{a}_k) s_{i,j} s_{k,j}, & \text{PPNM.} \end{cases} \tag{3}$$

In (1) and (3), $\boldsymbol{a}_i \in \mathbb{R}^{n\times 1}$ is the ith endmember vector whose abundance in $\boldsymbol{x}_j$ is $s_{i,j}$. n, m , and r denote the numbers of spectral bands, pixels, and endmembers, respectively, and $\boldsymbol{\varepsilon}_j$ is the additive noise. $\boldsymbol{x}_j^{1\text{st}}$ is the contribution of single scatterings, and $\boldsymbol{x}_j^{2\text{nd}}$ represents the contribution of the second-order scatterings. $\boldsymbol{a}_i \odot \boldsymbol{a}_k = (a_{i,1}a_{k,1}, \ldots, a_{i,n}a_{k,n})^{\mathrm{T}}$ is the Hadamard product between $\boldsymbol{a}_i$ and $\boldsymbol{a}_k$, which corresponds to the product of two material reflectances at each band. The products of abundances and specific nonlinear parameters are further also used for the adjustment of nonlinearity. The nonlinear parameter $\gamma_{i,k,j}$ of the GBM ranges from 0 to 1, and $\xi_j \in \mathbb{R}$ in the PPNM is also a nonlinear parameter. In these models, (4) should be satisfied to make abundances nonnegative and sum to one in every pixel

$$s_{i,j} \geq 0, \sum_{i=1}^{r} s_{i,j} = 1. \tag{4}$$

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B. SMLM

The MLM [12] extends the BMMs by including all degrees of multiple scatterings among endmembers. In pixel $\boldsymbol{x}_j$, a parameter $\underline{P}_j$ is used to represent the light's probability of undergoing further interactions. $(1-\underline{P}_j)$ means the probability of reaching the sensor. Therefore, the contributions of different scattering terms, including $\boldsymbol{a}_i$, $\boldsymbol{a}_i \odot \boldsymbol{a}_k$, $\boldsymbol{a}_i \odot \boldsymbol{a}_k \odot \boldsymbol{a}_l$, and so on, can be integrated using an expression similar to $\underline{P}_j$'s geometric distribution. Equation (5) describes an MLM-based pixel

$$\begin{aligned}\boldsymbol{x}_j &= \left(1-\underline{P}_j\right)\sum_{i=1}^{r}\boldsymbol{a}_i s_{i,j} \\ &\quad +\left(1-\underline{P}_j\right)\underline{P}_j\sum_{i=1}^{r}\sum_{k=1}^{r}(\boldsymbol{a}_i \odot \boldsymbol{a}_k)s_{i,j}s_{k,j} \\ &\quad +\left(1-\underline{P}_j\right)\underline{P}_j^2\sum_{i=1}^{r}\sum_{k=1}^{r}\sum_{l=1}^{r}(\boldsymbol{a}_i \odot \boldsymbol{a}_k \odot \boldsymbol{a}_l)s_{i,j}s_{k,j}s_{l,j} + \cdots + \boldsymbol{\varepsilon}_j \\ &= \frac{\left(1-\underline{P}_j\right)\sum_{i=1}^{r}\boldsymbol{a}_i s_{i,j}}{1-\underline{P}_j\sum_{i=1}^{r}\boldsymbol{a}_i s_{i,j}} + \boldsymbol{\varepsilon}_j. \end{aligned} \tag{5}$$

The SMLM is an extension of the MLM that considers the influence of shadows [46]. As shown in Fig. 1, SMLM is assumed to remove $(1-\underline{P}_j)\sum_{i=1}^{r}\boldsymbol{a}_i s_{i,j}$ from (5) for a fully shadowed pixel because the first-order scatterings are missing in the shadows. A shadow parameter $Q_j \in [0,\ 1]$ is introduced for the pixels containing partial shadows. The SMLM-based pixel can be described as follows:

$$\boldsymbol{x}_j = \frac{\left(1-\underline{P}_j\right)\sum_{i=1}^{r}\boldsymbol{a}_i s_{i,j}}{1-\underline{P}_j\sum_{i=1}^{r}\boldsymbol{a}_i s_{i,j}} - Q_j\left(1-\underline{P}_j\right)\sum_{i=1}^{r}\boldsymbol{a}_i s_{i,j} + \boldsymbol{\varepsilon}_j \tag{6}$$

where $Q_j = 0$ means that no shadows exist in $\boldsymbol{x}_j$, and $Q_j = 1$ corresponds to the situation in which $\boldsymbol{x}_j$ is fully shadowed.

C. Basic Global PSO

The design of PSO was initially inspired by birds' foraging behaviors [48]. PSO uses the heuristic movement of a group of particles in the feasible solution space to solve complex optimization problems. Every particle is considered to be a potential solution, which updates its position and velocity according to valuable swarm information. In the global PSO, all the particles learn from their own best positions in history and the best particle's position in the swarm. The entire swarm will move closer to the optimal solution after each update of the particle positions. Specifically, the update of the ith particle's position and velocity can be described by the following equation:

$$\begin{cases} \boldsymbol{V}_i^{(t+1)} = \omega\cdot\boldsymbol{V}_i^{(t)} + \underline{c}_1\cdot\underline{r}_1^{(t)} \\ \qquad\qquad \cdot\left(\boldsymbol{P}_{\text{ipbest}}^{(t)} - \boldsymbol{P}_i^{(t)}\right) + \underline{c}_2\cdot\underline{r}_2^{(t)}\cdot\left(\boldsymbol{P}_{\text{gbest}}^{(t)} - \boldsymbol{P}_i^{(t)}\right) \\ \boldsymbol{P}_i^{(t+1)} = \boldsymbol{P}_i^{(t)} + \boldsymbol{V}_i^{(t+1)} \end{cases} \tag{7}$$

where $\boldsymbol{P}_i^{(t)}$ and $\boldsymbol{V}_i^{(t)}$ are the ith particle's position and velocity in iteration t, respectively; their sizes are the same; $\boldsymbol{P}_{\text{ipbest}}^{(t)}$ is the best position of the ith particle in history; $\boldsymbol{P}_{\text{gbest}}^{(t)}$ is the swarm's global best position; and $\boldsymbol{P}_{\text{ipbest}}^{(t)}$ and $\boldsymbol{P}_{\text{gbest}}^{(t)}$ are determined according to the values of a fitness function $f(\cdot)$. A particle's position in a minimization problem is $\boldsymbol{P}_{\text{gbest}}^{(t)}$ if it provides the swarm's minimum value of the fitness function. $\boldsymbol{P}_{\text{ipbest}}^{(t+1)}$ is $\boldsymbol{P}_{\text{ipbest}}^{(t)}$ if $f(\boldsymbol{P}_{\text{ipbest}}^{(t)}) \le f(\boldsymbol{P}_{\text{ipbest}}^{(t+1)})$; otherwise, it equals $\boldsymbol{P}_i^{(t+1)}$. ω is an inertia weight. If ω is large, the swarm's exploration is dominant in the solution search, and if ω is small, the swarm turns to exploit finer positions. $\underline{c}_1$ and $\underline{c}_2$ are the acceleration coefficients, and $\underline{r}_1^{(t)}$ and $\underline{r}_2^{(t)}$ are uniformly drawn in the set (0, 1).

The multiple direction-based search mode makes it easier for PSOs to obtain better solutions. PSO does not require the objective functions to be differentiable nor gradient calculations to be performed. These characteristics bring PSO favorable conditions to address nonlinear unmixing problems as a generic optimization framework.

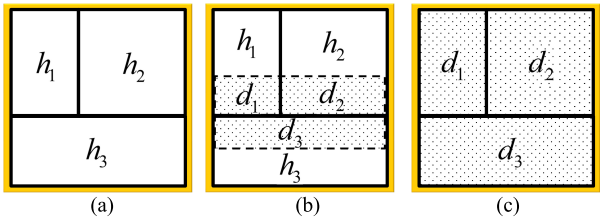


Fig. 2. Schematic of three endmembers' sunlit coefficients h_1, h_2, h_3 and shadowed coefficients d_1, d_2, d_3 in the same pixel under different illumination conditions. (a) Fully illuminated: each endmember's abundance s_i $(i = 1, 2, 3)$ is h_i. (b) Partially illuminated: each endmember's abundance s_i is $d_i + h_i$. (c) Fully shadowed: each endmember's abundance s_i is d_i.

III. Proposed Model and Unmixing Algorithm

A. Bilinear Mixture Model With Shadow Compensation

As shown in Fig. 1, for the same observed scenario, the number of fully illuminated pixels may be different as the spatial resolution changes. We hypothesize that a pixel may be partially or fully occupied by shadows to consider the shadowing effect in nonlinear unmixing. Accordingly, a pixel $\boldsymbol{x}_j$ can be directly divided into two parts: a sunlit part $\boldsymbol{x}_j^{\text{sunlit}}$ and a shadowed part $\boldsymbol{x}_j^{\text{shadow}}$. Thus, the endmember set $\boldsymbol{a}_1, \boldsymbol{a}_2, \ldots, \boldsymbol{a}_r$ describes these two parts based on a sunlit coefficient vector $\boldsymbol{h}_j = (h_{1,j},\ h_{2,j}, \ldots,\ h_{r,j})^{\mathrm{T}}$ and a shadowed coefficient vector $\boldsymbol{\Delta}_j = (d_{1,j},\ d_{2,j}, \ldots,\ d_{r,j})^{\mathrm{T}}$ of the jth pixel $\boldsymbol{x}_j$, respectively, which is shown in Fig. 2. It can be inferred that the abundance vector $\boldsymbol{s}_j$ equals $\boldsymbol{h}_j + \boldsymbol{\Delta}_j$.

Based on the illumination-driven separation of every pixel, an NMM that considers shadows can be built. The total irradiance reaching ground covers primarily comes from direct sunlight, the light scattered by the atmosphere, and the light reflected by the surroundings [28], [55]. In this study, the total irradiance is generalized to contain the direct irradiance $\boldsymbol{I}_{\text{direct}}$ and the indirect irradiance $\boldsymbol{I}_{\text{indirect}}$. A sunlit pixel receives both $\boldsymbol{I}_{\text{direct}}$ and $\boldsymbol{I}_{\text{indirect}}$. However, for a fully shadowed pixel, the illumination source is only the indirect irradiance $\boldsymbol{I}_{\text{indirect}}$, which is induced by the diffused skylight and the multiple scatterings from its surroundings. Therefore, the proportion

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of multiple scatterings increases markedly in shadows due to the absence of direct sunlight. However, since the illumination source is considered the same for all the pixels in the atmospheric correction, the objects' reflectance inside the shadows is smaller than their intrinsic reflectance.

According to the RT theory and the radiance $\boldsymbol{R}_{\text{reference}}$ of a calibration panel [27], [43], the reflectance of $\boldsymbol{x}_{\text{sunlit}}$ and $\boldsymbol{x}_{\text{shadow}}$ corresponding to a sunlit pixel and a fully shadowed pixel, respectively, can be simplified as

$$\begin{cases} \boldsymbol{x}_{\text{sunlit}} \leftarrow \dfrac{(\boldsymbol{I}_{\text{direct}} + \boldsymbol{I}_{\text{indirect}}) \odot \boldsymbol{x}_{\text{sunlit}}}{\boldsymbol{R}_{\text{reference}}} \\ \boldsymbol{x}_{\text{shadow}} \leftarrow \dfrac{\boldsymbol{I}_{\text{indirect}} \odot \boldsymbol{x}_{\text{shadow}}}{\boldsymbol{R}_{\text{reference}}} \end{cases} \tag{8}$$

where $\boldsymbol{I}_{\text{direct}}$, $\boldsymbol{I}_{\text{indirect}}$, and $\boldsymbol{R}_{\text{reference}}$ are wavelength-dependent. In practice, it is reasonable to rewrite $\boldsymbol{x}_{\text{sunlit}}$ in the form of the traditional BMM in (9). A relaxed expression of $\boldsymbol{x}_{\text{shadow}}$ can also be approximated by (10), which is a scaled version of $\boldsymbol{x}^{1\text{st}}$

$$\boldsymbol{x}_{\text{sunlit}} = \sum_{i=1}^{r} \boldsymbol{a}_i h_i + \sum_{i=1}^{r}\sum_{k=1}^{r} \boldsymbol{a}_i \odot \boldsymbol{a}_k h_i h_k \tag{9}$$

$$\boldsymbol{x}_{\text{shadow}} = \boldsymbol{c} \odot \sum_{i=1}^{r} \boldsymbol{a}_i d_i \tag{10}$$

where $\boldsymbol{c} \in \mathbb{R}^{n\times 1}$ is a wavelength-dependent vector whose values range from 0 to 1 and is defined to simply approximate the weak illumination in the shadows and the influence of scene-dependent prior physical parameters, such as the sky factor and illumination angles. $\boldsymbol{c}$ is assumed to be constant for the entire observed scene in this study. Consequently, by combining (9), (10), and the second-order scatterings between $\boldsymbol{x}_j^{\text{sunlit}}$ and $\boldsymbol{x}_j^{\text{shadow}}$, an SCBMM is proposed in (11) for every pixel. In addition, the constraints in (12) should be satisfied

$$\begin{aligned} \boldsymbol{x}_j = {} & \boldsymbol{x}_j^{\text{sunlit}} \sum_{i=1}^{r} h_{i,j} + \boldsymbol{x}_j^{\text{shadow}} \sum_{i=1}^{r} d_{i,j} \\ & + \boldsymbol{x}_j^{\text{sunlit}} \odot \boldsymbol{x}_j^{\text{shadow}} \left(\sum_{i=1}^{r} h_{i,j} \right) \left(\sum_{i=1}^{r} d_{i,j} \right) \end{aligned} \tag{11}$$

$$0 \le h_{i,j} \le s_{i,j}, \quad 0 \le d_{i,j} \le s_{i,j}, \quad \sum_{i=1}^{r} \left(h_{i,j} + d_{i,j} \right) = 1 \tag{12}$$

where $h_{i,j}$ and $d_{i,j}$ are the ith elements of $\boldsymbol{h}_j$ and $\boldsymbol{\Delta}_j$, respectively. They represent the sunlit coefficient and the shadowed coefficient of the ith endmember in the jth pixel, respectively. For a pixel $\boldsymbol{x}_j$, its sunlit part $\boldsymbol{x}_j^{\text{sunlit}}$ and its shadowed part $\boldsymbol{x}_j^{\text{shadow}}$ are defined by (9) and (10), respectively. It is noted that our model has removed nonlinear parameters, such as $\gamma_{i,k,j}$ and ξ_j of the GBM and PPNM, to ease the optimization and avoid the uncertainty induced by nonlinear parameters in the estimation of the shadowed coefficients. Hence, the flexibility and generalization of the proposed model may be affected.

Since the jth pixel's abundance vector is $\boldsymbol{s}_j = \boldsymbol{h}_j + \boldsymbol{\Delta}_j$, it can be concluded that a pixel $\boldsymbol{x}_j$ is represented by the traditional BMM in (9) if $\boldsymbol{h}_j = \boldsymbol{s}_j$ and $\boldsymbol{\Delta}_j = \boldsymbol{0}$; thus, pixel $\boldsymbol{x}_j$ is modeled as a fully shadowed pixel in (10) if $\boldsymbol{\Delta}_j = \boldsymbol{s}_j$ and $\boldsymbol{h}_j = \boldsymbol{0}$. Otherwise, $\boldsymbol{x}_j$ is defined by a high-level macroscopic BMM constructed by $\boldsymbol{x}_j^{\text{sunlit}}$ and $\boldsymbol{x}_j^{\text{shadow}}$. Compared with traditional BMMs and existing unmixing methods that consider shadows, the SCBMM interprets the nonlinearity and shadows concisely; there is no requirement to select shadowed endmembers. Scenario-based prior physical parameters and other auxiliary data, such as DSMs, are also not required. Therefore, this model has good potential to be used in practical nonlinear unmixing.

B. Constrained Optimization Problem for Unmixing

A compact optimization problem can be generated by substituting $\boldsymbol{h}_j = \boldsymbol{s}_j - \boldsymbol{\Delta}_j$ into (11) and minimizing the reconstruction error (RE) using the Euclidean distance as follows:

$$\begin{aligned} & \min_{\boldsymbol{S},\, \boldsymbol{\Delta},\, \boldsymbol{c}} f_{\text{SCBMM}} \\ & = \sum_{j=1}^{m} \left\| \begin{array}{l} \boldsymbol{x}_j^{\text{sunlit}} \left(1 - \widehat{d}_j\right) + \boldsymbol{x}_j^{\text{shadow}} \widehat{d}_j \\ + \boldsymbol{x}_j^{\text{sunlit}} \odot \boldsymbol{x}_j^{\text{shadow}} \left(1 - \widehat{d}_j\right) \widehat{d}_j - \boldsymbol{x}_j \end{array} \right\|_F^2 \\ & \text{subject to } 0 \le d_{i,j} \le s_{i,j} \le 1, \quad \sum_{i=1}^{r} s_{i,j} = 1, \ \boldsymbol{0}_n < \boldsymbol{c} < \boldsymbol{1}_n \end{aligned} \tag{13}$$

where $\boldsymbol{x}_j^{\text{sunlit}}$ and $\boldsymbol{x}_j^{\text{shadow}}$ are given in (9) and (10), respectively. $\widehat{d}_j$ is the summation of all the elements in $\boldsymbol{\Delta}_j$, i.e., $\widehat{d}_j = \sum_{i=1}^{r} d_{i,j}$. f_{SCBMM} denotes the RE based on the SCBMM. $\boldsymbol{0}_n$ and $\boldsymbol{1}_n$ are column vectors of all 0s and 1s, respectively. Thus, an abundance matrix $\boldsymbol{S}$, a shadowed coefficient matrix $\boldsymbol{\Delta}$, and a scaled vector $\boldsymbol{c}$ in (10) must be estimated for unmixing when the endmembers $\boldsymbol{a}_1, \boldsymbol{a}_2, \ldots, \boldsymbol{a}_r$ are known. However, (13) is nonconvex, and the boundary constraint of $\boldsymbol{\Delta}$ is dynamic in the optimization. Thus, calculating an accurate solution by directly solving such a high-dimensional problem remains a difficult task.

On the other hand, the presence of shadows in a hyperspectral image typically occurs in a few pixels, implying that $\boldsymbol{\Delta}$ must be sparse. Shadows are commonly observed on the boundary between different adjacent ground covers, and the neighboring pixels in a spatially local region often have similar abundances. Considering these inherent features, two regularization terms are added to (13), which helps to limit the solution space and avoid overfitting. Thus, abundances in line with ground truth have more chances to be estimated. The sparsity of $\boldsymbol{\Delta}$ is considered using the ℓ_1 norm [56], which has been widely used in the literature to improve unmixing performance [2]–[4]. In addition, the TV constraint [57] is enforced on $\boldsymbol{S}$ to obtain spatially smooth abundances and favor the identification of the shadows on the boundary. These two regularization terms have the following mathematical forms:

$$\left\| \boldsymbol{\Delta} \right\|_{1,1} = \sum_{j=1}^{m} \left\| \boldsymbol{\Delta}_j \right\|_1 = \sum_{j=1}^{m}\sum_{i=1}^{r} \left| d_{i,j} \right| \tag{14}$$

$$\left\| \boldsymbol{S} \right\|_{\text{TV}} = \sum_{j=1}^{m}\sum_{k=1}^{2} \left(\left| \boldsymbol{s}_j - \boldsymbol{s}_j^{v_k} \right| + \left| \boldsymbol{s}_j - \boldsymbol{s}_j^{h_k} \right| \right) \tag{15}$$

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where $\|\cdot\|_{1,1}$, $\|\cdot\|_{\mathrm{TV}}$, and $|\cdot|$ represent the ℓ_1 norm, the TV regularizer, and the absolute value operator, respectively; and $\boldsymbol{s}_j^v$ and $\boldsymbol{s}_j^h$ correspond to the abundance vectors of the pixels neighboring $\boldsymbol{x}_j$ in the vertical (i.e., top and bottom) and horizontal (i.e., left and right) directions, respectively. After regularizing (13) with (14) and (15), we can write the constrained optimization problem for nonlinear unmixing as

$$\begin{aligned} &\min_{\boldsymbol{S},\,\boldsymbol{\Delta},\,\boldsymbol{c}} f_{\mathrm{SCBMM}} + \lambda_1 \|\boldsymbol{S}\|_{\mathrm{TV}} + \lambda_2 \|\boldsymbol{\Delta}\|_{1,1} \\ &\text{subject to } 0 \le d_{i,j} \le s_{i,j} \le 1, \quad \sum_{i=1}^{r} s_{i,j} = 1, \ \mathbf{0}_n < \boldsymbol{c} < \mathbf{1}_n \end{aligned} \tag{16}$$

where λ_1 and λ_2 are two regularization parameters to control the influence of constraints. Their values markedly depend on the spatial features of different hyperspectral images.

C. Unmixing Based on Multiswarm PSO

1) Algorithm Principles: Constrained optimization problems, such as (16), are common in studies of unmixing algorithms, and many are solved using gradient-based numerical optimization methods. Thus, careful mathematical derivation for different variables of objective functions must be performed repeatedly for specific unmixing problems. Such an idea certainly encounters many obstacles in practical applications because the assumption of nonlinear unmixing gradually approaches actual complex situations. Unfortunately, the nonconvexity and nondifferentiability of objective functions could also hamper the access of one-way search-based gradient methods to acceptable local optima. The generation of reliable solutions relies on robust optimization methods; thus, it makes sense to build a versatile and user-friendly optimization framework to solve unmixing problems.

We, thus, estimate the unknown variables in (16) by resorting to PSO, which can easily provide solutions for complex objective functions. Once the optimization problem is formulated, a group of particles search for the swarm's best position (i.e., the optimal solution) via information exchange. Thus, the derivation of an objective function is no longer necessary, and regularizations can be directly integrated into the PSO framework. Also, boundary control for restricting unmixing variables' values to lie in a certain interval can be conveniently implemented. This fact implies that PSO has more chances to produce better unmixing results.

Since three variables $\boldsymbol{S}$, $\boldsymbol{\Delta}$, and $\boldsymbol{c}$ in (16) must be estimated, as shown in Fig. 3, three swarms named $\mathrm{Swarm}_{\boldsymbol{S}}$, $\mathrm{Swarm}_{\boldsymbol{\Delta}}$, and $\mathrm{Swarm}_{\boldsymbol{c}}$ are constructed to update $\boldsymbol{S}$, $\boldsymbol{\Delta}$, and $\boldsymbol{c}$, respectively. A multiswarm PSO algorithm is developed in which these three swarms evolute alternately. Each swarm is designed following the idea of global PSO in Section II-C. The reduced optimization subproblems for these swarms are written in (17). For example, $\mathrm{Swarm}_{\boldsymbol{S}}$ is responsible for the minimization with respect to $\boldsymbol{S}$ in (16). Without a loss of generality, a particle's position in $\mathrm{Swarm}_{\boldsymbol{S}}$ represents a potential abundance matrix $\boldsymbol{S}$ and can be updated using (7). In each iteration, the global best position $\boldsymbol{P}_{\boldsymbol{S}_\mathrm{gbest}}$ of $\mathrm{Swarm}_{\boldsymbol{S}}$ is delivered to $\mathrm{Swarm}_{\boldsymbol{\Delta}}$ and $\mathrm{Swarm}_{\boldsymbol{c}}$ as a known variable.

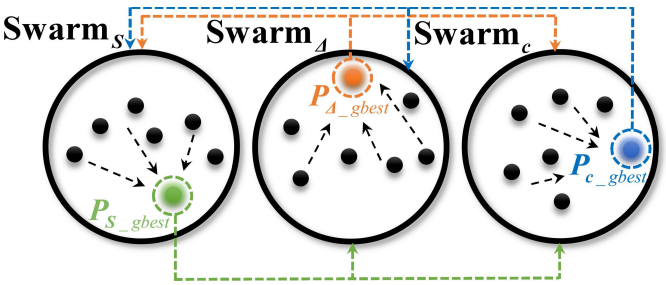


Fig. 3. Principle of PSO with multiple swarms for alternately updating the abundances $\boldsymbol{S}$, shadowed coefficients $\boldsymbol{\Delta}$, and scaled vector $\boldsymbol{c}$. The global best particle's position of each swarm is used to promote the evolution of the other two swarms.

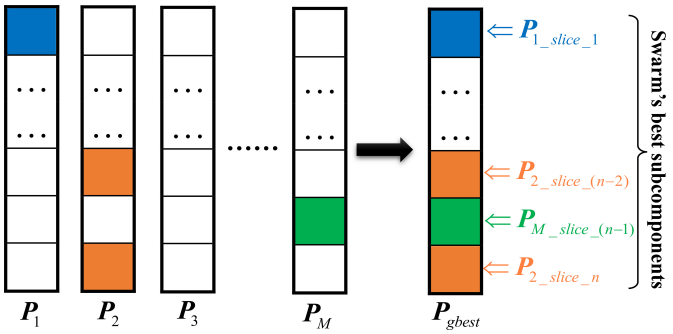


Fig. 4. Determination of the global best particle's position based on the idea of divide and conquer. According to the division based on the pixels or bands, the swarm's best subcomponents in every subdimension, which are supplied by different particles joined together into $\boldsymbol{P}_{\mathrm{gbest}}$.

Conversely, the global best positions of $\mathrm{Swarm}_{\boldsymbol{\Delta}}$ and $\mathrm{Swarm}_{\boldsymbol{c}}$, i.e., $\boldsymbol{P}_{\boldsymbol{\Delta}_\mathrm{gbest}}$ and $\boldsymbol{P}_{\boldsymbol{c}_\mathrm{gbest}}$, are used to determine the fitness of every particle in $\mathrm{Swarm}_{\boldsymbol{S}}$

$$\begin{cases} \min_{\boldsymbol{S}} f_{\mathrm{SCBMM}} + \lambda_1 \|\boldsymbol{S}\|_{\mathrm{TV}}, & 0 \le s_{i,j} \le 1, \ \sum_{i=1}^{r} s_{i,j} = 1 \\ \min_{\boldsymbol{\Delta}} f_{\mathrm{SCBMM}} + \lambda_2 \|\boldsymbol{\Delta}\|_{1,1}, & 0 \le d_{i,j} \le s_{i,j} \\ \min_{\boldsymbol{c}} f_{\mathrm{SCBMM}}, & \mathbf{0}_n < \boldsymbol{c} < \mathbf{1}_n. \end{cases} \tag{17}$$

However, although the basic PSO performs well when solving low-dimensional problems whose feasible solution spaces are small, the basic PSO suffers from the "curse of dimensionality" [47]. Unfortunately, the high dimensionality of hyperspectral data can induce the loss of population diversity and yield a premature issue to the basic PSO. The idea of dimensional division in our previous study [54] is used to overcome this drawback because the original high-dimensional unmixing problem in (16) is separable in this work. Specifically, f_{SCBMM} and regularization terms in (17) can be equivalently split into the sum of bandwise or pixelwise subitems. Such a property can strongly facilitate the dimensional division of (17) into a series of simple parallel subproblems. Thus, fitness functions can be vectorized, leading to the information exchange of all the particles' positions at finely divided subdimensions. Fig. 4 shows a brief illustration of the determination of $\boldsymbol{P}_{\mathrm{gbest}}$. The divide-and-conquer PSO allows the best position to be combined flexibly by multiple

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finely divided best subcomponents, promoting an increase in population diversity and unmixing accuracy.

Vectorized fitness functions $\boldsymbol{fitness}_{\boldsymbol{S}}$, $\boldsymbol{fitness}_{\boldsymbol{\Delta}}$, and $\boldsymbol{fitness}_{\boldsymbol{c}}$ of particles in Swarm$_{\boldsymbol{S}}$, Swarm$_{\boldsymbol{\Delta}}$, and Swarm$_{\boldsymbol{c}}$, respectively, are shown in (18), where $\boldsymbol{fitness}_{\boldsymbol{S}}$ and $\boldsymbol{fitness}_{\boldsymbol{\Delta}}$ are obtained through the pixel-based division, and $\boldsymbol{fitness}_{\boldsymbol{c}}$ follows the bandwise division

$$\begin{cases} \boldsymbol{fitness}_{\boldsymbol{S}} \\ \quad = \left(f_{\text{SCBMM}}^{\text{C1}}+\lambda_1 g_1,\ f_{\text{SCBMM}}^{\text{C2}}+\lambda_1 g_2,\ \ldots,\ f_{\text{SCBMM}}^{\text{C}m}+\lambda_1 g_m\right) \\ \boldsymbol{fitness}_{\boldsymbol{\Delta}} \\ \quad = \left(f_{\text{SCBMM}}^{\text{C1}}+\lambda_2 z_1,\ f_{\text{SCBMM}}^{\text{C2}}+\lambda_2 z_2,\ \ldots,\ f_{\text{SCBMM}}^{\text{C}m}+\lambda_2 z_m\right) \\ \boldsymbol{fitness}_{\boldsymbol{c}} \\ \quad = \left(f_{\text{SCBMM}}^{\text{R1}},\ f_{\text{SCBMM}}^{\text{R2}},\ \ldots,\ f_{\text{SCBMM}}^{\text{R}n}\right)^{\text{T}} \end{cases} \tag{18}$$

$$\begin{cases} f_{\text{SCBMM}}^{\text{C}j} = \left\| \begin{array}{l} \boldsymbol{x}_j^{\text{sunlit}}\left(1-\widehat{d}_j\right)+\boldsymbol{x}_j^{\text{shadow}}\widehat{d}_j+ \\ \boldsymbol{x}_j^{\text{sunlit}}\odot\boldsymbol{x}_j^{\text{shadow}}\left(1-\widehat{d}_j\right)\widehat{d}_j-\boldsymbol{x}_j \end{array} \right\|_2^2 \\ f_{\text{SCBMM}}^{\text{R}k} = \left\| \begin{array}{l} \boldsymbol{X}_{k,:}^{\text{sunlit}}\odot\left(\boldsymbol{1}_m^{\text{T}}-\widehat{\boldsymbol{d}}^{\text{T}}\right)+\boldsymbol{X}_{k,:}^{\text{shadow}}\odot\widehat{\boldsymbol{d}}^{\text{T}} \\ +\boldsymbol{X}_{k,:}^{\text{sunlit}}\odot\boldsymbol{X}_{k,:}^{\text{shadow}}\left(\boldsymbol{1}_m^{\text{T}}-\widehat{\boldsymbol{d}}^{\text{T}}\right)\odot\widehat{\boldsymbol{d}}^{\text{T}}-\boldsymbol{X}_{k,:} \end{array} \right\|_2^2 . \end{cases} \tag{19}$$

Algorithm 1 NUSPSO

Input: Hyperspectral data $\boldsymbol{X} \in \mathbb{R}^{n\times m}$, endmember matrix $\boldsymbol{A}$.

Output: Abundance matrix $\boldsymbol{S} \in \mathbb{R}^{r\times m}$ (i.e., $\boldsymbol{P}_{\boldsymbol{S}_gbest}$), shadow parameter matrix $\boldsymbol{\Delta} \in \mathbb{R}^{r\times m}$ (i.e., $\boldsymbol{P}_{\boldsymbol{\Delta}_gbest}$), and scaled vector $\boldsymbol{c} \in \mathbb{R}^{n\times 1}$ (i.e., $\boldsymbol{P}_{\boldsymbol{c}_gbest}$).

Set $t = 0$. **Generate** three swarms, i.e., Swarm$_{\boldsymbol{S}}$, Swarm$_{\boldsymbol{\Delta}}$, and Swarm$_{\boldsymbol{c}}$, and initialize their particles' positions $\boldsymbol{P}_i^{(0)}$ and velocities $\boldsymbol{V}_i^{(0)}$.

While $t < t_{\max}$ **do**

a) Calculate the particles' fitness vectors of Swarm$_{\boldsymbol{S}}$ using $\boldsymbol{P}_{\boldsymbol{\Delta}_gbest}^{(t)}$ and $\boldsymbol{P}_{\boldsymbol{c}_gbest}^{(t)}$ according to (18). Determine $\boldsymbol{P}_{\boldsymbol{S}_ipbest}^{(t+1)}$ and $\boldsymbol{P}_{\boldsymbol{S}_gbest}^{(t+1)}$ by (20). Update $\boldsymbol{P}_i^{(t+1)}$ and $\boldsymbol{V}_i^{(t+1)}$ in Swarm$_{\boldsymbol{S}}$ with (7).

b) Calculate the particles' fitness vectors of Swarm$_{\boldsymbol{\Delta}}$ using $\boldsymbol{P}_{\boldsymbol{S}_gbest}^{(t+1)}$ and $\boldsymbol{P}_{\boldsymbol{c}_gbest}^{(t)}$ according to (18). Determine $\boldsymbol{P}_{\boldsymbol{\Delta}_ipbest}^{(t+1)}$ and $\boldsymbol{P}_{\boldsymbol{\Delta}_gbest}^{(t+1)}$ by (20). Update $\boldsymbol{P}_i^{(t+1)}$ and $\boldsymbol{V}_i^{(t+1)}$ in Swarm$_{\boldsymbol{\Delta}}$ with (7).

c) Calculate the particles' fitness vectors of Swarm$_{\boldsymbol{c}}$ using $\boldsymbol{P}_{\boldsymbol{S}_gbest}^{(t+1)}$ and $\boldsymbol{P}_{\boldsymbol{\Delta}_gbest}^{(t+1)}$ according to (18). Determine $\boldsymbol{P}_{\boldsymbol{c}_ipbest}^{(t+1)}$ and $\boldsymbol{P}_{\boldsymbol{c}_gbest}^{(t+1)}$ by (20). Update $\boldsymbol{P}_i^{(t+1)}$ and $\boldsymbol{V}_i^{(t+1)}$ in Swarm$_{\boldsymbol{c}}$ with (7).

d) $t = t + 1$.

End

In (18), $f_{\text{SCBMM}}^{\text{C}j}$ and $f_{\text{SCBMM}}^{\text{R}k}$ ($j = 1, 2, \ldots, m$, $k = 1, 2, \ldots, n$) are defined in (19). $f_{\text{SCBMM}}^{\text{C}j}$ is f_{SCBMM}'s jth subcomponent obtained through pixel-based division. $f_{\text{SCBMM}}^{\text{R}k}$ is f_{SCBMM}'s kth subcomponent obtained through the band-based division. $g_j = \sum_{i=1}^{r}\sum_{k=1}^{2}\left(|s_{i,j}-s_{i,j}^{v_k}|+|s_{i,j}-s_{i,j}^{h_k}|\right)$ and $z_j = \sum_{i=1}^{r}|d_{i,j}|$ denote the jth subcomponents of the ℓ_1 norm and the TV regularizer based on the pixel-based division, respectively. In (19), $\boldsymbol{X}_{k,:}$ is the kth row vector of the hyperspectral data matrix $\boldsymbol{X}$ and $\widehat{\boldsymbol{d}} = (\widehat{d}_1, \widehat{d}_2, \ldots, \widehat{d}_m)^{\text{T}}$ whose jth element $\widehat{d}_j$ is given in (13).

Finally, in comparison of every particle's fitness vectors according to (18), the jth subcomponent slices of $\boldsymbol{P}_{\text{pbest}}^{(t+1)}$ and $\boldsymbol{P}_{\text{gbest}}^{(t+1)}$ in a swarm containing M particles in the tth iteration can be determined as follows:

$$\begin{cases} \boldsymbol{P}_{\text{pbest_slice_}j}^{(t+1)} \\ \quad = \begin{cases} \boldsymbol{P}_{\text{pbest_slice_}j}^{(t)}, & \text{fitness}\left(\boldsymbol{P}_{\text{pbest}}^{(t)}\right)_j \le \text{fitness}\left(\boldsymbol{P}_{\text{pbest}}^{(t+1)}\right)_j \\ \boldsymbol{P}_{\text{pbest_slice_}j}^{(t+1)}, & \text{fitness}\left(\boldsymbol{P}_{\text{pbest}}^{(t)}\right)_j > \text{fitness}\left(\boldsymbol{P}_{\text{pbest}}^{(t+1)}\right)_j \end{cases} \\ \boldsymbol{P}_{\text{gbest_slice_}j}^{(t+1)} = \boldsymbol{P}_{i_\text{slice_}j}^{(t)} \end{cases}$$
$$\text{fitness}\left(\boldsymbol{P}_i^{(t)}\right)_j = \min\left(\text{fitness}\left(\boldsymbol{P}_1^{(t)}\right)_j, \ldots, \text{fitness}\left(\boldsymbol{P}_M^{(t)}\right)_j\right) \tag{20}$$

where $\text{fitness}(\boldsymbol{P}_{\text{pbest}}^{(t)})_j$ and $\text{fitness}(\boldsymbol{P}_i^{(t)})_j$ represent the jth fitness scalar value of the fitness vectors of $\boldsymbol{P}_{\text{pbest}}^{(t)}$ and $\boldsymbol{P}_i^{(t)}$, respectively, calculated by (18) for swarms. $\boldsymbol{P}_{\text{pbest_slice_}j}^{(t)}$ and $\boldsymbol{P}_{\text{gbest_slice_}j}^{(t)}$ are the jth subcomponent slices of $\boldsymbol{P}_{\text{pbest}}^{(t)}$ and $\boldsymbol{P}_{\text{gbest}}^{(t)}$, respectively. $\boldsymbol{P}_{i_\text{slice_}j}^{(t)}$ denotes the jth subcomponent slice of $\boldsymbol{P}_i^{(t)}$, and $\text{fitness}(\boldsymbol{P}_i^{(t)})_j$ is the best in a swarm. During the iterations, (7) updates the particle positions in Swarm$_{\boldsymbol{S}}$, Swarm$_{\boldsymbol{\Delta}}$, and Swarm$_{\boldsymbol{c}}$.

2) Boundary Control and PSO Parameter Settings: The boundary conditions in (17) can be satisfied by adjusting each iteration's particle positions. Let $\boldsymbol{B}_{\min}$ and $\boldsymbol{B}_{\max}$ denote the variable's lower bound matrix and upper bound matrix, respectively. A damping and absorbing strategy [54] in (21) is improved for boundary control in this study

$$\begin{cases} \boldsymbol{P}_i^{(t+1)} = \boldsymbol{P}_i^{(t)} - \text{rand}_{0-1}\cdot\left(\boldsymbol{B}_{\max}-\boldsymbol{P}_i^{(t)}\right) & \boldsymbol{P}_i^{(t+1)} > \boldsymbol{B}_{\max} \\ \boldsymbol{P}_i^{(t+1)} = \boldsymbol{P}_i^{(t)} + \text{rand}_{0-1}\cdot\left(\boldsymbol{B}_{\min}-\boldsymbol{P}_i^{(t)}\right) & \boldsymbol{P}_i^{(t+1)} < \boldsymbol{B}_{\min} \end{cases} \tag{21}$$

where rand_{0-1} is a random number drawn from (0, 1). All elements in $\boldsymbol{B}_{\min}$ are equal to zero; $\boldsymbol{B}_{\max}$ is filled with ones in the cases of $\boldsymbol{S}$ and $\boldsymbol{c}$; and $\boldsymbol{B}_{\max}$ of $\boldsymbol{\Delta}$ is set dynamically as the Swarm$_{\boldsymbol{S}}$'s global best position $\boldsymbol{P}_{\boldsymbol{S}_\text{gbest}}^{(t+1)}$. Accordingly, the particle velocities should comply with the following restrictions:

$$\boldsymbol{V}_i^{(t)} = \min\left(\max\left(\boldsymbol{V}_i^{(t)}, -\boldsymbol{B}_{\max}\right), \boldsymbol{B}_{\max}\right) \tag{22}$$

where the elements of $\boldsymbol{V}_i^{(t)}$ are set as zero if the elements of $\boldsymbol{P}_i^{(t)}$ at the same positions violate the boundary constraints.

In addition, the positions of Swarm$_{\boldsymbol{S}}$'s particles are normalized in each iteration to satisfy the sum-to-one constraint. The inertia weight ω and acceleration coefficients c_1 and c_2 change dynamically during the iterations according to (23) and (24) to improve the stability and efficiency of PSO

$$\omega^{(t)} = \omega_{\max} - (\omega_{\max}-\omega_{\min})t/t_{\max}, \quad \omega^{(t)} \in [0.4, 0.95] \tag{23}$$
$$\underline{c}_1^{(t)} = 2.5 - t/t_{\max}, \quad \underline{c}_1^{(t)} = 1.5 + t/t_{\max} \tag{24}$$

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where $t_{\max}$ is the maximum number of iterations. Finally, the proposed supervised nonlinear spectral unmixing method with shadow compensation using multiswarm PSO (NUSPSO) can be briefly summarized in Algorithm 1. With the use of given endmembers, NUSPSO is designed for abundance estimation. Methods such as the vertex component analysis (VCA) [58] can be used for endmember extraction. The velocities of all the particles are initialized to be 0s, and the particle positions of $\text{Swarm}_{\boldsymbol{S}}$ and $\text{Swarm}_{\boldsymbol{c}}$ are randomly initialized in the interval [0, 1]. Abundances estimated by FCLS are added into $\text{Swarm}_{\boldsymbol{S}}$ as a particle's initial position to facilitate convergence. The initial global best positions $\boldsymbol{P}^{(0)}_{\text{gbest}}$ of the three swarms are obtained by cross calculation of every particle's fitness according to (18). The global best position $\boldsymbol{P}^{(0)}_{\boldsymbol{S}_\text{gbest}}$ of $\text{Swarm}_{\boldsymbol{S}}$ is used to randomly initialize the particles' positions of $\text{Swarm}_{\boldsymbol{\Delta}}$ as an upper bound matrix. In this study, the swarm size M is set as 20, and the maximum number of iterations $t_{\max}$ is 200. In each iteration, the three swarms evolute alternately and provide their own global best positions to the other swarms until the stopping criteria are satisfied.

IV. Experiments and Discussion

In this section, three different types of experiments are performed to validate the proposed algorithm. NUSPSO's robustness to the number of endmembers, the noise, the accuracy of endmembers, and the percentage of shadowed pixels are first analyzed in the experiments with basic model-based simulated data. A physical-based simulated dataset of a virtual citrus orchard and three real hyperspectral remote sensing images are also considered to complement the sound evaluation of the proposed algorithm's practical performance in the presence of real shadows.

Because this study focuses on abundance estimation, endmembers are assumed to be known in advance. NUSPSO is compared with a supervised linear unmixing method FCLS [20], two BMM-based nonlinear abundance estimation algorithms [9], [10] via GDA (i.e., GBM-GDA and PPNM-GDA), and two supervised nonlinear unmixing methods based on the MLM [12] and the SMLM [46] considering the shadowing effect in these experiments. In addition, two shadow removal methods based on unmixing are also considered. The first [41] unmixes the shadowed pixels based on an extended LMM and alternating optimization, and is referred to as SR-LU in this article. The second [45] combines extracted endmembers from the shadowed regions with a kernel-based nonlinear unmixing method K-Hype [14] to unmix the shadows and is referred to as SR-KHype in this article. These methods rely on the prior determination of shadowed regions, and sunlit pixels are unmixed by their corresponding unmixing methods and sunlit endmembers.

The corresponding studies of existing methods are considered to set parameter values. The maximum iteration number was 3000, and the tolerance was 10^{-8} for GBM-GDA and PPNM-GDA. The MLM and the SMLM were optimized via sequential quadratic programming where two tolerance values were set as 10^{-8} and 10^{-10}, respectively. The shadowed pixels for SR-LU and SR-KHype were identified according to the shadowed coefficients of the simulated data. In terms of real hyperspectral data, a matched filter and a thresholding method in [31] were used to determine the shadowed areas for these two methods. To develop a quantitative and qualitative experimental understanding of the proposed algorithm and compared methods, the root mean square error (RMSE) defined in (25) and RE defined in (26) were used to quantitatively measure the unmixing accuracy

$$\text{RMSE} = \sqrt{\frac{1}{mr}\sum_{j=1}^{m}\left\|\boldsymbol{s}_j - \hat{\boldsymbol{s}}_j\right\|_2^2} \tag{25}$$

$$\text{RE} = \sqrt{\frac{1}{mn}\sum_{j=1}^{m}\left\|\boldsymbol{x}_j - \hat{\boldsymbol{x}}_j\right\|_2^2} \tag{26}$$

where $\boldsymbol{s}_j$ is the true abundance vector of the observed pixel $\boldsymbol{x}_j$, and $\hat{\boldsymbol{s}}_j$ is $\boldsymbol{x}_j$'s estimated abundance vector. $\hat{\boldsymbol{x}}_j$ in (26) is a model-based reconstructed pixel. In contrast to the RMSE of abundances, the RE of data typically serves as an auxiliary reference in the literature due to possible overfitting [24]. Abundance maps generated by these algorithms and shadowed coefficient maps also provide visual comparisons. All the compared algorithms were executed in ten independent Monte Carlo runs, and the averages and standard deviations of the RMSEs and REs of the unmixing results were compared. The experiments were performed on a computer with a 3.7-GHz Intel[1] Core[2] i9 CPU and 64 GB of memory.

A. Experimental Results With Simulated Data

Simulated datasets were generated according to the SCBMM for the first type of experiment. In data generation, six endmembers corresponding to spectra of ground covers, such as plants and minerals, were selected from the United States Geological Survey (USGS) digital spectral library.[3] Each endmember contains 224 spectral bands covering wavelengths from 380 to 2500 nm. The Gaussian random fieldmethod provided in the Hyperspectral Imagery Synthesis toolbox[4] was applied for abundance map generation. Because shadows typically exist on the boundary between different adjacent ground covers, the abundance maps' horizontal and vertical spatial gradients were calculated. Then, a certain number of pixels with large absolute values of gradients were regarded as possible shadowed pixels. Their shadow coefficients $\boldsymbol{\Delta}$ corresponding to each endmember and a scaled vector $\boldsymbol{c}$ were randomly produced. Additive white Gaussian noise was finally added to the pure data according to different signal-to-noise ratios [$\text{SNR} = 10\log_{10}(E[\boldsymbol{x}^{\mathrm{T}}\boldsymbol{x}]/E[\boldsymbol{\varepsilon}^{\mathrm{T}}\boldsymbol{\varepsilon}])$] [2]. Except for the last experiment with regard to comparing running time, the simulated data used in the other experiments consist of 64×64 pixels. In the following experiments, the influence of endmembers, noise, and the percentage of shadowed pixels are discussed. Finally, the computational

[1] Registered trademark.

[2] Trademarked.

[3] Available at http://speclab.cr.usgs.gov/spectral-lib.html

[4] Available at http://www.ehu.es/ccwintco/index.php/Hyperspectral_Imagery_Synthesis_tools_for_MATLAB

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TABLE I
COMPARISON OF UNMIXING ACCURACIES OF ALL ALGORITHMS WITH DIFFERENT ENDMEMBERS' NUMBERS

Endmembers' numbers		FCLS	GBM-GDA	PPNM-GDA	MLM	SR-LU	SR-KHype	SMLM	NUSPSO
3	RMSE	0.2062 ± 0.0006	0.2120 ± 0.0000	0.1189 ± 0.0020	0.1770 ± 0.0001	0.2266 ± 0.0000	0.1419 ± 0.0013	0.1746 ± 0.0001	**0.0638 ± 0.0012**
	RE	0.1996 ± 0.0008	0.2013 ± 0.0005	0.0212 ± 0.0043	0.1572 ± 0.0006	0.2115 ± 0.0005	0.0270 ± 0.0054	0.1571 ± 0.0006	**0.0201 ± 0.0048**
4	RMSE	0.2566 ± 0.0007	0.2636 ± 0.0002	0.1161 ± 0.0010	0.2059 ± 0.0004	0.2841 ± 0.0000	0.1694 ± 0.0091	0.1956 ± 0.0009	**0.0766 ± 0.0006**
	RE	0.2496 ± 0.0009	0.2713 ± 0.0005	**0.0227 ± 0.0054**	0.1993 ± 0.0006	0.2613 ± 0.0005	0.0336 ± 0.0068	0.1993 ± 0.0006	0.0230 ± 0.0054
5	RMSE	0.2925 ± 0.0008	0.3046 ± 0.0000	0.1133 ± 0.0009	0.2484 ± 0.0003	0.2962 ± 0.0000	0.1645 ± 0.0078	0.2360 ± 0.0004	**0.0889 ± 0.0006**
	RE	0.1918 ± 0.0008	0.1976 ± 0.0005	0.0206 ± 0.0047	0.1374 ± 0.0007	0.1684 ± 0.0006	0.0308 ± 0.0059	0.1376 ± 0.0007	**0.0200 ± 0.0050**
6	RMSE	0.3095 ± 0.0006	0.3014 ± 0.0001	0.1320 ± 0.0012	0.1771 ± 0.0003	0.3033 ± 0.0001	0.1468 ± 0.0065	0.1448 ± 0.0013	**0.1070 ± 0.0005**
	RE	0.2358 ± 0.0007	0.2374 ± 0.0004	0.0214 ± 0.0047	0.1842 ± 0.0006	0.1947 ± 0.0005	0.0324 ± 0.0072	0.1845 ± 0.0005	**0.0206 ± 0.0049**

TABLE II
COMPARISON OF UNMIXING ACCURACIES OF ALL ALGORITHMS WITH DIFFERENT TYPES OF ENDMEMBERS

Percentage of shadow pixels		FCLS	GBM-GDA	PPNM-GDA	MLM	SR-LU	SR-KHype	SMLM	NUSPSO
True	RMSE	0.2566 ± 0.0007	0.2636 ± 0.0002	0.1161 ± 0.0010	0.2059 ± 0.0004	0.2841 ± 0.0000	0.1694 ± 0.0091	0.1956 ± 0.0009	**0.0766 ± 0.0006**
	RE	0.2496 ± 0.0009	0.2713 ± 0.0005	**0.0227 ± 0.0054**	0.1993 ± 0.0006	0.2613 ± 0.0005	0.0336 ± 0.0068	0.1993 ± 0.0006	0.0230 ± 0.0054
VCA	RMSE	0.2477 ± 0.0654	0.2490 ± 0.0652	0.1416 ± 0.0907	0.1975 ± 0.1049	0.2679 ± 0.0564	0.1664 ± 0.0977	0.1949 ± 0.1086	**0.0889 ± 0.0011**
	RE	0.2031 ± 0.0028	0.2169 ± 0.0055	0.0316 ± 0.0040	0.1908 ± 0.0030	0.2004 ± 0.0030	**0.0204 ± 0.0064**	0.1906 ± 0.0030	0.0339 ± 0.0037
Graph-geodesic DmaxD	RMSE	0.4616 ± 0.2093	0.3860 ± 0.0935	0.3228 ± 0.1086	0.3624 ± 0.1010	0.3802 ± 0.0907	0.3597 ± 0.1330	0.3631 ± 0.1069	**0.1921 ± 0.0870**
	RE	0.2194 ± 0.0119	0.2476 ± 0.0120	0.1062 ± 0.0267	0.2085 ± 0.0139	0.2288 ± 0.0127	**0.0584 ± 0.0298**	0.2083 ± 0.0139	0.0858 ± 0.0325

complexity of NUSPSO is analyzed, and the execution time of the algorithms is compared. The parameters λ_1 and λ_2 of NUSPSO are related to the data's spatial structural properties, which were, respectively, set as 0.2 and 0.8 in this section through cross-validation.

1) Analysis of Sensitivity to Endmembers: Since the compared algorithms are all supervised, the influence of endmembers, including their amount and accuracy on abundance estimation, is discussed first. The SNR is 40 dB, and 10% of the data are shadowed pixels. Table I lists the RMSEs and REs obtained by these algorithms in detail when the number of endmembers changes from 3 to 6, and true endmembers are used. The best values are marked in bold. More endmembers are shown to make it more difficult to obtain accurate unmixing results with all the algorithms, which is reasonable because more explanatory variables tend to make regression more complex. With the number of endmembers fixed to 4, Table II also highlights the algorithms' sensitivities to the accuracy of endmembers. True endmembers and endmembers extracted by VCA and a nonlinear distance-MaxD (DMaxD) algorithm [21] were used in the experiments. DMaxD used a nonlinear manifold-based graph-geodesic distance in this article, implying that this method can be used for nonlinear endmember extraction compared to the linear VCA. The parameter K of the DMaxD's K-nearest neighbor graph was set as 20. This table shows that the endmembers' errors can decrease the unmixing accuracy of NUSPSO. However, due to its good interpretability for the simulated data, NUSPSO performs better than the other compared methods even when incorrect endmembers are used.

In Tables I and II, the overall unmixing accuracies of seven compared traditional or state-of-the-art algorithms are much worse than NUSPSO. The complexity of simulated data, thus, likely plays an important role in deteriorating the unmixing results. Most of the algorithms fail to estimate accurate abundances and suffer from the severe comprehensive influence induced by shadows and nonlinear mixing effects. PPNM-GDA and SR-KHype yield better abundances than the other five compared methods due to their good generalization. Since the proposed method's fitting ability can be affected by removing nonlinear parameters, these two algorithms' REs can outperform NUSPSOs in some cases. However, it seems that the increase in NUSPSO's RE does not affect its abundance estimation, indicating that small REs do not always mean good unmixing accuracy. Due to its interpretability with shadows and nonlinearity, and its good optimization strategy, NUSPSO yields the best unmixing results under nearly all situations and exhibits considerable robustness to the effects of endmembers in these experiments.

2) Noise Robustness Analysis: The presence of noise typically reduces the unmixing accuracy. The following experiments discuss each algorithm's robustness to noise using data with SNR being 50, 40, 30, and 20 dB. The number of endmembers is four, and the percentage of shadowed pixels is 10% for each endmember. The comparison of unmixing accuracies can be observed in Table III. NUSPSO provides the best unmixing results among all the algorithms, even when the dataset is heavily corrupted by noise. Fig. 5 shows a visual illustration of the abundance maps produced by the algorithms. In contrast to the ground truth, NUSPSO's abundance maps show much more similarities than the others. The first four algorithms that inherently ignore the presence of shadows incorrectly recognize endmembers in the shadows, resulting in the misestimation of abundances. The contributions from nonlinear mixing effects and shadows seem to confuse nonlinear unmixing algorithms, such as GBMGDA and PPNMGDA. With SR-LU, although it considers the shadows in linear unmixing, the nonlinear mixing effects seem to seriously affect its estimated abundances compared to the last three algorithms that deal with the nonlinear mixing effects and shadows simultaneously. The SMLM mitigates this issue to a certain degree by introducing a shadow parameter into the MLM,

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TABLE III
COMPARISON OF UNMIXING ACCURACIES OF ALL ALGORITHMS WITH DIFFERENT SNRS

SNRs		FCLS	GBM-GDA	PPNM-GDA	MLM	SR-LU	SR-KHype	SMLM	NUSPSO
50 dB	RMSE	0.2558 ± 0.0001	0.2630 ± 0.0001	0.1162 ± 0.0007	0.2056 ± 0.0003	0.2841 ± 0.0000	0.1559 ± 0.0017	0.1947 ± 0.0003	**0.0763 ± 0.0004**
	RE	0.2480 ± 0.0001	0.2705 ± 0.0000	**0.0116 ± 0.0010**	0.1983 ± 0.0001	0.2605 ± 0.0000	0.0197 ± 0.0013	0.1983 ± 0.0001	0.0123 ± 0.0011
40 dB	RMSE	0.2566 ± 0.0007	0.2636 ± 0.0002	0.1161 ± 0.0010	0.2059 ± 0.0004	0.2841 ± 0.0000	0.1694 ± 0.0091	0.1956 ± 0.0009	**0.0766 ± 0.0006**
	RE	0.2496 ± 0.0009	0.2713 ± 0.0005	**0.0227 ± 0.0054**	0.1993 ± 0.0006	0.2613 ± 0.0005	0.0336 ± 0.0068	0.1993 ± 0.0006	0.0230 ± 0.0054
30 dB	RMSE	0.2640 ± 0.0031	0.2631 ± 0.0006	0.1170 ± 0.0017	0.2098 ± 0.0021	0.2840 ± 0.0001	0.1636 ± 0.0052	0.2039 ± 0.0034	**0.0769 ± 0.0009**
	RE	0.2608 ± 0.0062	0.2789 ± 0.0045	0.0656 ± 0.0195	0.2094 ± 0.0059	0.2692 ± 0.0047	0.0759 ± 0.0188	0.2093 ± 0.0059	**0.0654 ± 0.0195**
20 dB	RMSE	0.2787 ± 0.0037	0.2689 ± 0.0030	0.1346 ± 0.0064	0.2272 ± 0.0061	0.2841 ± 0.0003	0.1919 ± 0.0227	0.2279 ± 0.0079	**0.0916 ± 0.0067**
	RE	0.3464 ± 0.0378	0.3544 ± 0.0353	0.2235 ± 0.0565	0.3015 ± 0.0414	0.3470 ± 0.0362	0.2254 ± 0.0540	0.3014 ± 0.0414	**0.2231 ± 0.0565**

TABLE IV
COMPARISON OF UNMIXING ACCURACIES OF ALL ALGORITHMS WITH DIFFERENT PERCENTAGE OF SHADOWED PIXELS

Percentage of shadow pixels		FCLS	GBM-GDA	PPNM-GDA	MLM	SR-LU	SR-KHype	SMLM	NUSPSO
5%	RMSE	0.2669 ± 0.0008	0.2681 ± 0.0004	0.0834 ± 0.0004	0.2222 ± 0.0006	0.2953 ± 0.0000	0.1458 ± 0.0037	0.2139 ± 0.0008	**0.0606 ± 0.0003**
	RE	0.2628 ± 0.0009	0.2830 ± 0.0005	**0.0229 ± 0.0060**	0.2096 ± 0.0006	0.2766 ± 0.0005	0.0290 ± 0.0063	0.2096 ± 0.0006	0.0238 ± 0.0057
10%	RMSE	0.2566 ± 0.0007	0.2636 ± 0.0002	0.1161 ± 0.0010	0.2059 ± 0.0004	0.2841 ± 0.0000	0.1694 ± 0.0091	0.1956 ± 0.0009	**0.0766 ± 0.0006**
	RE	0.2496 ± 0.0009	0.2713 ± 0.0005	**0.0227 ± 0.0054**	0.1993 ± 0.0006	0.2613 ± 0.0005	0.0336 ± 0.0068	0.1993 ± 0.0006	0.0230 ± 0.0054
15%	RMSE	0.2523 ± 0.0006	0.2634 ± 0.0002	0.1434 ± 0.0007	0.1947 ± 0.0005	0.2808 ± 0.0000	0.1765 ± 0.0154	0.1835 ± 0.0010	**0.0888 ± 0.0005**
	RE	0.2388 ± 0.0008	0.2629 ± 0.0004	0.0230 ± 0.0049	0.1918 ± 0.0006	0.2511 ± 0.0004	0.0336 ± 0.0065	0.1917 ± 0.0006	**0.0221 ± 0.0052**
20%	RMSE	0.2527 ± 0.0005	0.2696 ± 0.0002	0.1689 ± 0.0012	0.1850 ± 0.0009	0.2838 ± 0.0000	0.1772 ± 0.0180	0.1724 ± 0.0013	**0.0970 ± 0.0006**
	RE	0.2281 ± 0.0008	0.2550 ± 0.0004	0.0230 ± 0.0044	0.1837 ± 0.0006	0.2385 ± 0.0004	0.0345 ± 0.0068	0.1836 ± 0.0006	**0.0212 ± 0.0050**

which can be clearly found in the second column of Fig. 5, where some shadowed pixels correctly disappear. However, abundances obtained by the SMLM in some regions are overestimated or underestimated. Conversely, the abundance maps of NUSPSO are similar to the ground truth. According to the unmixing results in Table III and Fig. 5, the proposed method can accurately recover the pixel abundances affected by shadows.

To evaluate the shadow identification ability, we also compare the shadow coefficients calculated by the SMLM and NUSPSO with the ground truth in Fig. 6. Specifically, the shadow coefficients of the latter two are the sum of shadow coefficients corresponding to every endmember. The spatial distribution of the shadowed coefficients of NUSPSO in Fig. 6(b) shows more similarities to the ground truth in Fig. 6(c). Compared with the SMLM, NUSPSO recognizes the most partially shadowed pixels with a low intensity of shadows. The experimental results in this section show that the proposed method achieves acceptable robustness to noise and can provide a reasonable explanation for the nonlinearity and shadows simultaneously in unmixing.

3) Analysis of Sensitivity to the Percentage of Shadowed Pixels: This experiment also describes the influence of shadowed pixel percentages. The percentage of shadowed pixels related to each endmember is set as 5%, 10%, 15%, and 20%. In Table IV, the unmixing accuracy of NUSPSO decreases marginally as more shadowed pixels occur in the data. Similar situations can be found in the results of algorithms such as PPNM-GDA and SR-KHype. The most accurate RMSEs of abundances are provided by the proposed method. PPNM-GDA also achieved accurate abundances near those of NUSPSO when the percentage of shadowed pixels equaled 5%, indicating that the presence of shadows may be a critical obstacle for traditional nonlinear unmixing methods to improve unmixing accuracy. In contrast, RMSEs and REs of the MLM and SMLM decrease when more shadowed pixels exist. Thus, their flexible expression for high-order nonlinearity can explain the constructed shadowed pixels in the simulated data.

4) Analysis of Computational Complexity and Running Time: The proposed method NUSPSO in Algorithm 1 implements unmixing through PSO with $\text{Swarm}_{\boldsymbol{S}}$, $\text{Swarm}_{\boldsymbol{\Delta}}$, and $\text{Swarm}_{\boldsymbol{c}}$. Its major computational costs come from the floating-point calculation and comparison of particle fitness values and the update of particle positions in these swarms. Let n, m, and r be the number of a hyperspectral image's spectral bands, pixels, and endmembers, respectively, and each swarm contains M particles. In terms of calculating and comparing the particles' fitness values, the worst case complexity of $\text{Swarm}_{\boldsymbol{S}}$ and $\text{Swarm}_{\boldsymbol{\Delta}}$ is $\mathbf{O}(r^2nmM + mM^2)$, and $\text{Swarm}_{\boldsymbol{c}}$'s is $\mathbf{O}(nmM + nM^2)$ in each iteration. In terms of updating the particles, the computational complexity of $\text{Swarm}_{\boldsymbol{S}}$ and $\text{Swarm}_{\boldsymbol{\Delta}}$ is $\mathbf{O}(rmM)$, and that of $\text{Swarm}_{\boldsymbol{c}}$ is $\mathbf{O}(nM)$. Therefore, the total computational complexity of NUSPSO in one iteration is approximately $\mathbf{O}(r^2nmM + (m+n)M^2)$.

Finally, the running time of eight compared algorithms is shown in Table V. Different numbers of endmembers and pixels were used to generate simulated data for the experiments. Aside from GBM-GDA and PPNM-GDA, most algorithms require no additional computation time as the number of endmembers increases. However, the change in these algorithms' computation time is strongly affected by the data size. The proposed method always requires more time due to its optimization mechanism of multiswarm PSO. However, in the study of nonlinear unmixing, the unmixing accuracy usually seems to be more critical. In light of multiswarm PSO's robust search ability and easy use, it can be considered that its computational burden may be acceptable, especially in dealing with complex nonconvex nonlinear unmixing problems. On the other hand, the intrinsic multiple direction-based search mode

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TABLE V
COMPARISON OF EXECUTION TIME (IN SECONDS) OF ALL ALGORITHMS WITH DIFFERENT ENDMEMBERS' NUMBERS AND PIXELS' SIZES

Execution time		FCLS	GBM-GDA	PPNM-GDA	MLM	SR-LU	SR-KHype	SMLM	NUSPSO
Endmembers' numbers	3	0.26	14.63	11.78	25.94	1.01	6.32	28.61	188.72
	4	0.40	13.40	62.71	27.09	1.58	6.13	30.07	190.75
	5	0.45	87.33	129.81	32.11	2.04	6.23	36.22	191.68
	6	0.57	191.01	159.24	33.00	3.09	5.13	38.57	192.75
Pixels' sizes	32×32	0.12	5.88	15.93	7.28	0.48	1.54	7.87	72.81
	64×64	0.40	13.40	62.71	27.09	1.58	6.13	30.07	190.75
	96×96	0.83	44.69	142.41	65.91	3.44	13.58	73.40	362.10
	128×128	1.56	60.22	244.23	107.39	4.90	24.74	120.38	635.72

of PSO enables the algorithm to be easily combined with parallel computing techniques to improve the computational efficiency when using NUSPSO.

B. Experimental Results With Virtual Orchard Data

In the experiments with model-based simulated data, the proposed algorithm's nonlinear unmixing performance and its ability to interpret shadows were validated. Although NUSPSO has demonstrated its superiority in handling ideal data compared to other algorithms, its performance should be further verified in real situations quantitatively. Thus, a physical-based simulated dataset of a virtual citrus orchard in Fig. 7 is considered in this study. This half-real dataset is generated by an extended physically based ray-tracing (PBRT) model and calibrated by *in situ* data. The available realistic light transmission paths and multiple scatterings have made it suitable for evaluating nonlinear unmixing in the literature [16]–[19], [24], [59]. A detailed description of the modules constituting the observed scene, such as the illumination sources and sensor platforms, can be found in [17]. Fig. 7(a) shows the original high-resolution image of the virtual citrus orchard, which consists of three endmembers, including soil, weed, and tree. These endmembers' spectral curves are drawn in Fig. 7(b). The dataset used for unmixing contains 20×20 pixels whose spectra are continuously sampled from 350 to 2500 nm. The water vapor absorption bands (99–110 and 143–161) were removed in the experiment.

As shown in Fig. 8, in addition to the multiple scatterings, shadows also exist in this virtual citrus orchard. Three endmembers are affected by shadows to different degrees. In particular, many pixels of soil and trees are covered by shadows. Fig. 9(a) also provides a quantitative illustration of the contribution of shadows in this dataset. The highest abundance of shadows is more than 0.4, and the total shadow contribution in all pixels is 33.12%. Only if the shadows are correctly removed, the real abundances of the three materials can be obtained accurately. Therefore, using this dataset to study the nonlinear mixing effects and shadows simultaneously is reliable.

Since the spatial structural characteristics of this dataset are not prominent and are different from the model-based simulated data, λ_1 and λ_2 of NUSPSO are set as 0.001 and 0.01, respectively, in this experiment. The detailed unmixing accuracies are provided in Table VI. The RMSE of the abundances estimated by SR-KHype is the smallest in the first seven algorithms, demonstrating its good fitting ability with a small RE. Although the RE of NUSPSO is larger than that of SR-KHype, NUSPSO achieves the best RMSE of abundances among all the algorithms. This experiment with virtual orchard data further shows that the proposed algorithm can implement nonlinear unmixing efficiently in the presence of shadows.

TABLE VI
COMPARISON OF UNMIXING ACCURACIES OF ALL ALGORITHMS WITH VIRTUAL CITRUS ORCHARD DATA

	FCLS	GBM-GDA	PPNM-GDA	MLM	SR-LU	SR-KHype	SMLM	NUSPSO
RMSE	0.1784	0.1560	0.1615	0.1902	0.1789	0.1543	0.1902	**0.1120**
RE	0.0253	0.0174	0.0121	0.0182	0.0073	**0.0027**	0.0182	0.0094

C. Experimental Results With Real Hyperspectral Images

The unmixing performance of the compared algorithms is finally evaluated with experiments based on real hyperspectral remote sensing imagery. Three widely used hyperspectral datasets with different spatial resolutions were used to perform these experiments. Shadows can be directly observed from these datasets or be assumed to exist inside some pixels. The possibility of no direct illumination in the pixels of two images and their subregions is provided quantitatively in Fig. 9(b) and (c) using a scaled shadow function-based matched filter method [31]. VCA was used to obtain the endmembers in advance for the abundance estimation executed by the above algorithms. In terms of experimental discussion, although many studies have used these data for unmixing evaluation, which can offer partial references for comparison, the real quantitative abundances of the ground truth are unknown. Thus, the RMSE of the estimated abundances is inaccessible, and the calculation of RE seems to be valueless. In this section, abundance maps generated by the compared algorithms are displayed for visual qualitative analysis. Because SMLM and NUSPSO account for the shadows in nonlinear unmixing, the total shadow coefficient maps provided by them are also used to discuss the identification of shadows. The referenced shadowed pixels used for SR-LU and SR-KHype were identified using the matched filter [31] and the thresholding method.

As shown in Fig. 10, the first real hyperspectral remote sensing image containing 614×512 pixels was acquired over Moffett Field, CA, USA, by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS) in 1997. A total of 224 spectral bands were continuously sampled in wavelength intervals from 0.4 to 2.5 μm. A subregion containing 50×50 pixels

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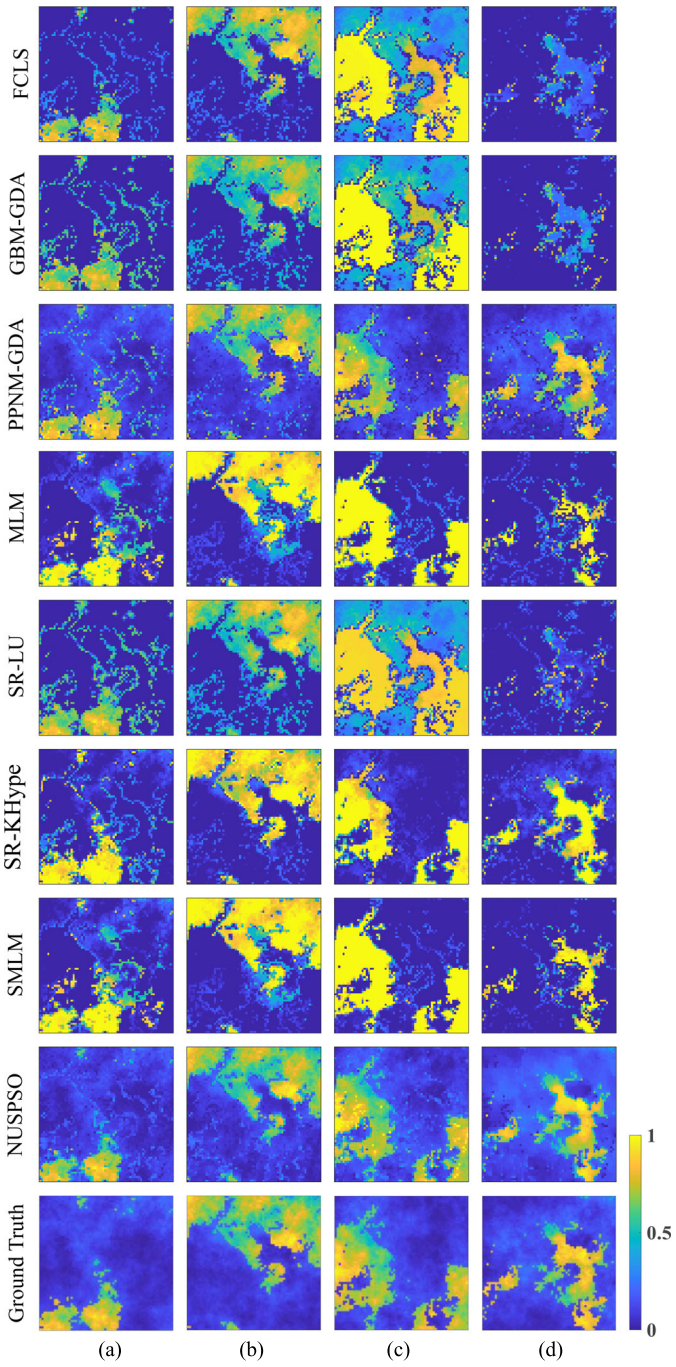


Fig. 5. Abundance maps estimated by the compared algorithms for the simulated dataset constructed by four endmembers: (a) Maple_Leaves DW92-1, (b) Olivine GDS70.a Fo89 165um, (c) Calcite CO2004, (d) Quartz GDS74 Sand Ottawa. The SNR is 40 dB, and the percentage of shadowed pixels is 10% for each endmember.

was selected where three major types of ground cover, including vegetation, water, and soil, are distributed. According to Fig. 9(b), the contributions of shadows in the entire image and this selected subregion were estimated to be 4.46% and 6.11%, respectively. This dataset's 20-m spatial resolution makes shadows difficult to observe directly by human eyes and likely to exist inside some pixels, and the contribution of shadows might be underestimated due to the low spatial

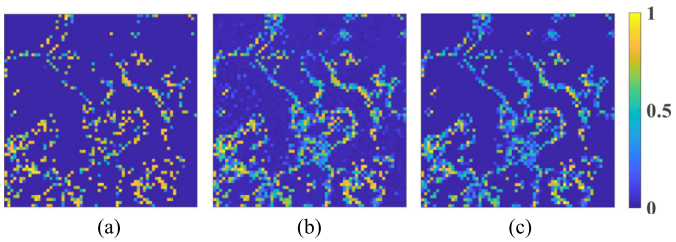


Fig. 6. Total distribution of shadow coefficients. (a) SMLM. (b) NUSPSO. (c) Ground Truth.

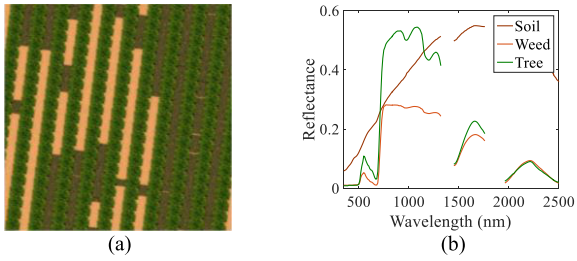


Fig. 7. Virtual citrus orchard data. (a) High-resolution scene image. (b) Spectral curves of three endmembers, including tree, weed, and soil.

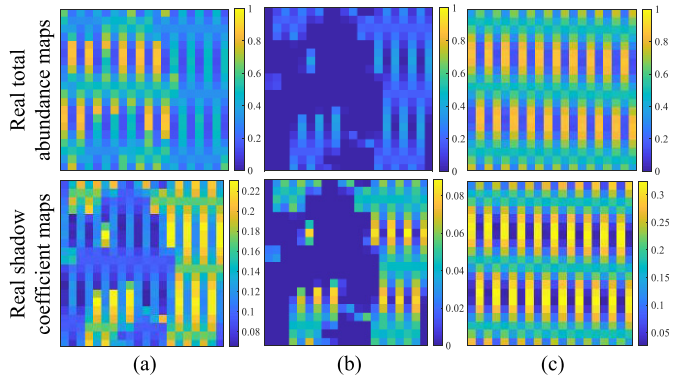


Fig. 8. Real abundance maps and real shadow coefficient maps of three endmembers in the virtual orchard dataset. (a) Soil. (b) Weed. (c) Tree.

resolution. Therefore, the shadows were not regarded as a specific type of endmember, but it is still meaningful to address the issue of shadows in nonlinear unmixing to recognize the real ground covers located in some special areas of this dataset. In the experiments, we removed the image's water vapor absorption and low SNR bands, and used 185 bands (5–104, 116–149, and 171–221). λ_1 and λ_2 of NUSPSO were set as 0.5 and 0.8, respectively.

Abundance maps of this dataset estimated by the algorithms are compared in Fig. 11. Three ground covers have been recognized correctly by most algorithms, and NUSPSO produces smoother abundance maps, possibly due to the TV spatial regularizations. We can, thus, conclude that the proposed algorithm can obtain satisfactory unmixing results for real hyperspectral images. In addition, Fig. 12 compares the shadow coefficient maps of the SMLM and the proposed method with the shadowed pixels identified by the matched filter method. The SMLM incorrectly identifies water as shadows because water and shadows have similar low spectral responses, which

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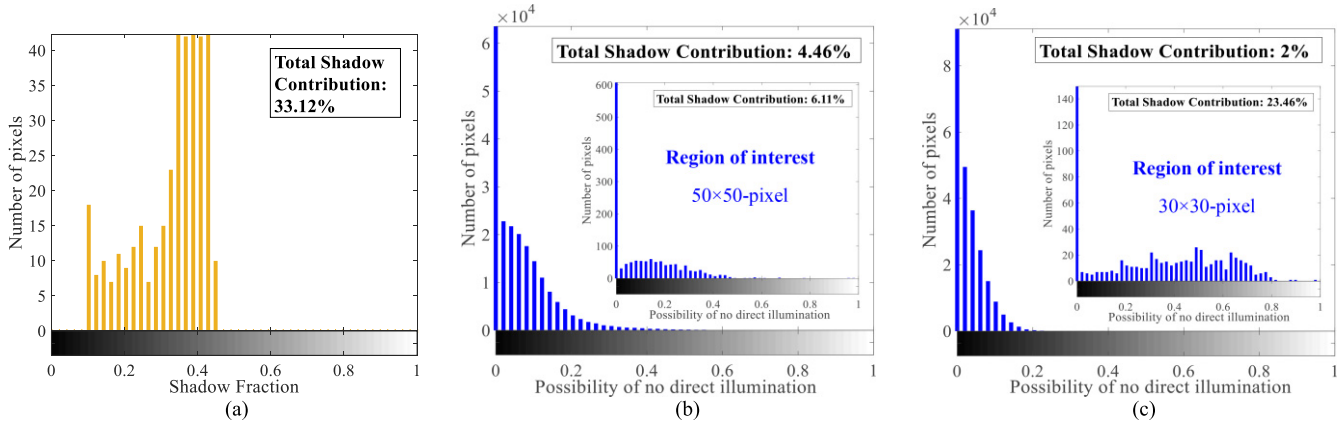


Fig. 9. Comparison of the shadows' contribution in different experimental datasets. (a) Real statistical distribution of shadow fractions in the virtual orchard dataset. Statistical distribution of the possibility of no direct illumination provided by a matched filter for (b) AVIRIS dataset and its subregion, and (c) HYDICE dataset and its subregion.



Fig. 10. Selected subregion of the AVIRIS Moffett Field dataset for experiments.

may confuse the algorithm. Similar issues have been reported in other studies [27]. In terms of the total shadow coefficient maps shown in this figure, NUSPSO seems to achieve better performance in describing the real distribution of the shadows than SMLM. Referring to related nonlinear unmixing results in the literature [9], [10], [24], the shadowed coefficient map of NUSPSO is consistent with the distribution of nonlinearity, which reflects the intimate connection between shadows and nonlinear mixing effects.

The second real hyperspectral image with the higher spatial resolution was acquired by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor over Washington DC Mall.[5] The original dataset has 210 bands covering wavelengths from 0.4 to 2.5 μm and consists of 307 $\times$ 1280 pixels. After noisy bands were removed, 165 bands (16–101, 108–136, and 153–202) were used in the experiments. Fig. 13 shows a selected subimage with a size of 30 $\times$ 30 pixels for algorithm evaluation. This subregion is primarily occupied by grasses, roads, and trees. Different from the first AVIRIS dataset, shadows can be clearly observed in this image of interest. Fig. 9(c) shows that, although the total percentage of pixels that are affected by shadows is approximately 2% in the entire dataset, the contribution of shadows has increased to more than 20% in the selected small subregion. Therefore, the impact of shadows on abundance estimation cannot be ignored. In addition, light's multiple scatterings between different objects are considered to reside in this vegetated area. The existence of nonlinear mixing effects and shadows enables this dataset to be used as a good paradigm to study the influence of shadows on nonlinear unmixing. Since the shadow endmember can be extracted, a comparative trial different from the previous experiment can be performed. The shadows were regarded as a specific endmember in the abundance estimation of FCLS, GBM-GDA, PPNM-GDA, and MLM. SMLM and NUSPSO only used three endmembers corresponding to real ground covers for unmixing. λ_1 and λ_2 of NUSPSO are 0.05 and 0.8.

In Fig. 14, abundance maps of grasses, roads, and trees are presented. In Fig.15, the shadow abundance maps produced by four sunlit-pixel-based unmixing algorithms are also compared with the referenced shadowed areas identified by the matched filer method and the total shadow coefficient maps estimated by SMLM and the proposed method. With the abundance maps, SMLM and NUSPSO describe clearer distributions of trees and roads than the other four algorithms. Explicitly, NUSPSO describes a majority of trees in the shadows in more detail. However, similar to SMLM, a small part of NUSPSO's road abundance map is marginally interfered with by shadows because these two objects have similar spectral shapes. Conversely, the total shadow coefficient maps estimated by SMLM and NUSPSO appear identical to the shadow abundance maps generated by the other four algorithms and the referenced shadowed areas. Thus, it can be inferred that the proposed algorithm can generally deal with the impact of shadows well during nonlinear unmixing.

The third real hyperspectral image is the Pavia University dataset[6] captured by the ROSIS sensor [46]. The geometric resolution is 1.3 m, and 103 bands were collected in the range of 430–838 nm with a spectral resolution of 4 nm. Nine classes of materials including shadows widely exist in the 610 $\times$ 340 pixel dataset. True labels of the class training pixels are

[5]Available at https://engineering.purdue.edu/~biehl/MultiSpec/hyperspectral.html

[6]Available at http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes#Pavia_University_scene

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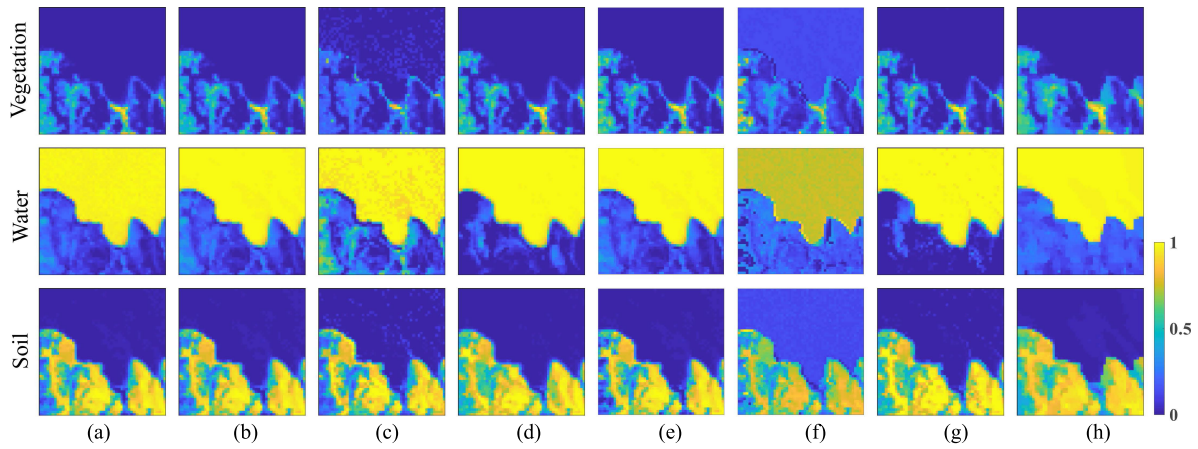


Fig. 11. Abundance maps produced by every algorithm for the AVIRIS data. (a) FCLS. (b) GBM-GDA. (c) PPNM-GDA. (d) MLM. (e) SR-LU. (f) SR-KHype. (g) SMLM. (h) NUSPSO.

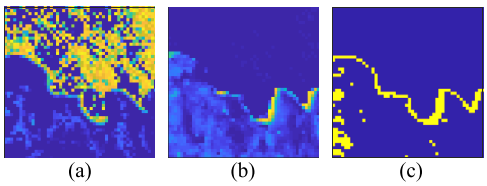


Fig. 12. Comparison of total shadow coefficient maps generated by (a) SMLM and (b) NUSPSO for AVIRIS data and (c) shadows identified by the matched filer.

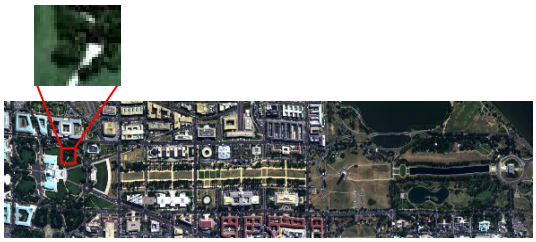


Fig. 13. Selected subregion of HYDICE data for experiments.

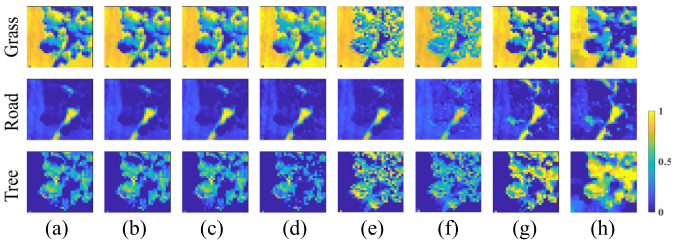


Fig. 14. Abundance maps obtained by each algorithm for HYDICE data. (a) FCLS. (b) GBM-GDA. (c) PPNM-GDA. (d) MLM. (e) SR-LU. (f) SR-KHype. (g) SMLM. (h) NUSPSO.

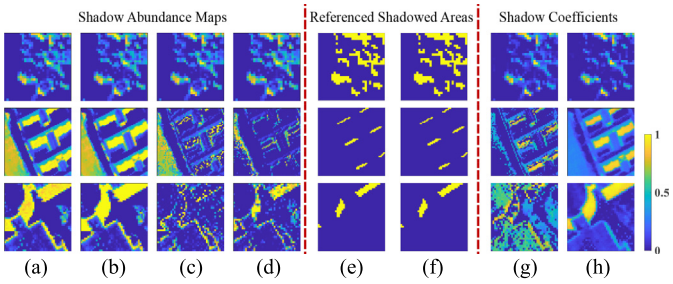


Fig. 15. Shadow abundance maps obtained by four algorithms. (a) FCLS. (b) GBM-GDA. (c) PPNM-GDA. (d) MLM. Referenced shadowed areas used by (e) SR-LU and (f) SR-KHype. Total shadow coefficient maps generated by (g) SMLM and (h) NUSPSO. From the top row to the bottom row, HYDICE data and two subregions of Pavia university data (Pavia #1 and Pavia #2).

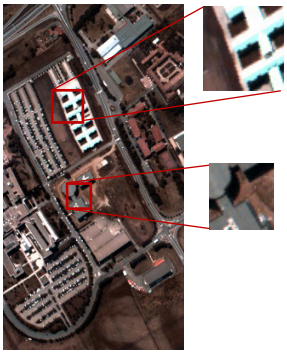


Fig. 16. Selected subregions of Pavia university data for experiments.

available. As shown in Fig. 16, we selected two subregions (i.e., Pavia #1 containing 60 $\times$ 60 pixels and Pavia #2 containing 40 $\times$ 40 pixels) of the Pavia University dataset for the experiments. Three main endmembers constitute these two subregions where shadows can be obviously observed. Each endmember's vector is obtained by averaging its corresponding labeled training pixels [46]. According to the class labels of shadows, there are 106 shadow pixels in Pavia #1 dataset and 83 shadow pixels in Pavia #2 dataset. The compared algorithms were performed like the experiments with HYDICE data. λ_1 and λ_2 of NUSPSO are 0.05 and 0.8.

Figs. 17 and 18 show three real endmembers' abundance maps estimated by the algorithms for Pavia #1 and Pavia #2, respectively. The shadow coefficient maps of SMLM and the proposed method are compared with referenced shadowed areas and the other algorithms' shadow abundance maps in

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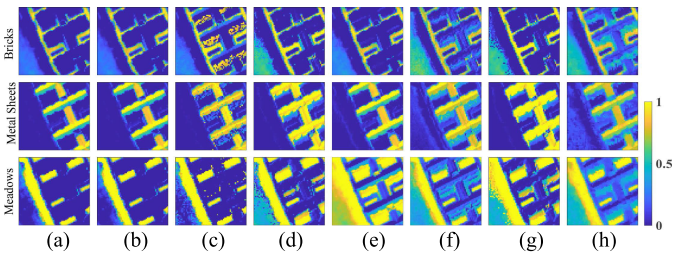


Fig. 17. Abundance maps obtained by each algorithm for the first subregion of Pavia university data (i.e., Pavia #1). (a) FCLS. (b) GBM-GDA. (c) PPNM-GDA. (d) MLM. (e) SR-LU. (f) SR-KHype. (g) SMLM. (h) NUSPSO.

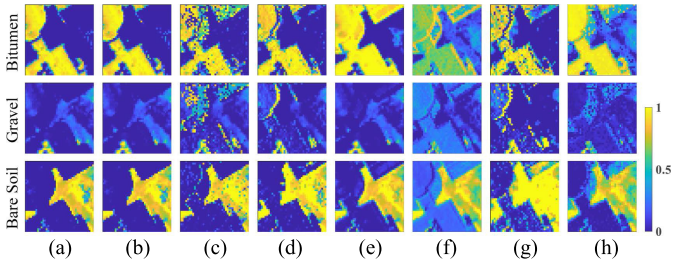


Fig. 18. Abundance maps obtained by each algorithm for the second subregion of Pavia university data (i.e., Pavia #2). (a) FCLS. (b) GBM-GDA. (c) PPNM-GDA. (d) MLM. (e) SR-LU. (f) SR-KHype. (g) SMLM. (h) NUSPSO.

TABLE VII
COMPARISON OF REs OF ALL ALGORITHMS WITH REAL HYPERSPECTRAL DATA

Real data	FCLS	GBM-GDA	PPNM-GDA	MLM	SR-LU	SR-KHype	SMLM	NUSPSO
Moffet field	0.0125	0.0119	*0.0098*	0.0102	0.0119	**0.0077**	*0.0098*	0.0164
HYDICE	0.0372	0.0376	0.0160	0.0151	0.0196	**0.0072**	*0.0147*	0.0186
Pavia #1	0.0543	0.0549	0.0213	0.0221	0.0587	**0.0081**	0.0221	*0.0191*
Pavia #2	0.0406	0.0427	0.0130	0.0122	0.0413	**0.0087**	0.0118	*0.0114*

Fig. 15. From these experimental results, we can see that NUSPSO can obtain reasonable abundance maps without using shadows as an endmember, and real ground covers inside shadows can be revealed to a certain degree. Particularly, the shadow coefficient maps in Fig. 15(h) generated by NUSPSO are in line with the referenced shadowed areas provided by shadows' true class labels in Fig. 15(e) and (f). It implies that NUSPSO can correctly recognize shadows to improve the nonlinear unmixing. Finally, the REs of the compared algorithms for real hyperspectral data are listed in Table VII. SR-KHype always has the smallest REs, and NUSPSO's REs can be small for the Pavia University dataset when shadows are apparent. However, considering SR-KHype's abnormal abundance maps, such as Figs. 11(f) and 18(f), it may suffer from the issue of overfitting. These experimental results further validate that the decrease in RE may not necessarily mean better abundances.

V. CONCLUSION

In this article, we presented a novel supervised spectral unmixing method that considers nonlinear mixing effects and shadows simultaneously. An extended BMM was constructed to organically integrate the pixels' sunlit and possibly shadowed components into a compact expression. Based on this model, two regularization terms that were properly enforced on abundances and shadow coefficients were used to formulate a new optimization problem for unmixing. Thus, the typical spatial structures of abundances and shadow coefficients can improve unmixing results. Finally, a user-friendly multiswarm PSO algorithm using dimensional division was extended and improved to solve this problem and achieve accurate abundances, which can be easily generalized to address complex nonlinear unmixing problems. Experimental results of the model-based simulated data, an open virtual orchard dataset, and two real hyperspectral images have validated that the proposed method can manage shadows with nonlinear unmixing and produce more accurate unmixing results compared with traditional and state-of-the-art unmixing algorithms considering shadows.

In future research, the relationship between shadows and spectral variability should be investigated. A valid method can be developed to provide interpretable discrimination of shadows and materials with low reflectances, such as water, in nonlinear unmixing. Moreover, it is meaningful to use parallel computing architectures, such as FPGA and GPU, to accelerate the proposed method.

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